From EMBER to CINDER: Sharpen the Concept, Burn It Once in Eight Languages, Spare the Word

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Abstract

[Abstract to be written after the experiments section is complete.]

1 Introduction

As language models (LMs) are deployed in more applications, the ability to erase specific knowledge from them becomes important for safety and compliance (Eldan and Russinovich, 2023; Li et al., 2024). A prominent approach is *persistent* concept erasure, which removes knowledge of a target concept by modifying the model's weights (Gur-Arieh et al., 2025; Ashuach et al., 2026). Recently, Suslik et al. (2026) showed that such methods overlook a part of the model that stores knowledge: the token embeddings. Their method, EMBER, decomposes the embeddings of concept-related tokens into sparse features, selects the features of the target concept with an LLM judge, and subtracts them from a small set of tokens.[1] Combined with MLP-based methods, it substantially improves robustness to relearning.

However, an embedding edit can only remove what its features capture, and how reliably EMBER finds a concept's features has not been studied. Its authors note that embeddings may not be an effective target for some concepts, that the tokens selected for editing may include semantically unrelated ones, and that the edit only applies to English tokens (Suslik et al., 2026, Limitations).

In this work, we hypothesize that these limitations stem in part from the geometry in which EMBER searches for features. A concept is carried by rare tokens, and in the raw embedding space rare tokens share a large common component, along the mean of all embeddings. EMBER's factorization therefore groups the concept with other rare tokens: the Harry Potter feature also collects

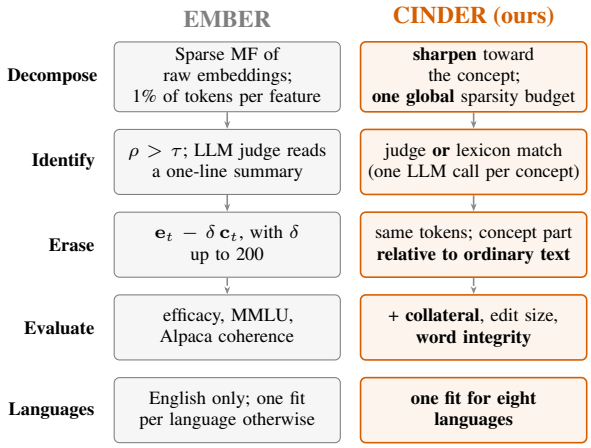


Figure 1: EMBER erases a concept by decomposing token embeddings into sparse features, identifying the concept's features, and subtracting them from concept tokens. CINDER keeps this framework and changes how the concept is found, verified and edited, so that quiet concepts become the strongest direction of the matrix. One CINDER fit serves eight languages, where EMBER needs a fit per language. We also extend the evaluation so that destroying the edited words no longer counts as erasing the concept.

Nestlé, Lego and Trump. Once this shared component is removed, the concept's tokens form a clean cluster, but a *quiet* one, of ordinary value to a factorization that minimizes reconstruction error, so it is often not found at all (Figure 2). Two more design choices compound the problem. Every feature keeps the same 1% of the tokens regardless of how broad its concept is, and the judge accepts a feature based on a one-sentence summary that often abstracts the concept away. When no feature is accepted, nothing is erased.

To tackle this gap, we introduce CINDER (Contrastive INformed Directional Embedding Removal; Figure 1), which keeps EMBER's localize-then-edit framework and changes how the concept is found. CINDER whitens the embeddings against ordinary text, which removes the shared

[1] Our code is available at https://github.com/guretskiysemyon/nlp_proj.

component, and sharpens them toward the concept's page, which makes the now clean concept cluster loud. It shares one sparsity budget across all features, so that each feature takes as many tokens as the data supports, and it verifies features both bottom-up, with EMBER's judge, and top-down, against a concept lexicon generated once per concept. Because the sharpened space is centred on ordinary text, CINDER's edit removes only what distinguishes the concept; an uncentred variant, CINDER-z, also removes part of the component that all tokens share, as EMBER's edit does.

Sharpening also makes multilingual erasure practical, which EMBER leaves to future work (Suslik et al., 2026, Limitations). Because EMBER edits the rows of English tokens, a concept it erases in English largely survives in other languages: on Gemma, edited models retain 0.19–0.24 more of the concept's knowledge in German, Spanish, French and Russian than in English. Running EMBER separately in every language multiplies its factorizations and judge calls by the number of languages, and fitting it once on multilingual text mostly fails: it accepts a concept feature for only 10 of 38 concepts. In a multilingual matrix the concept is quieter still, since language identity carries far more energy than the concept does, and EMBER's judge rejects most of the concept features that the fit does contain. CINDER whitens against a background with equal text in every language, which weakens the language directions, sharpens toward the concept's pages in all languages, and verifies features with its lexicon as well. A single CINDER fit over eight languages, with the uncentred edit of CINDER-z, then erases as well as eight separate EMBER runs, with one factorization and one judging pass instead of eight, and an eighth of the judge calls; to our knowledge, this is the first multilingual erasure at the embedding level.

We assess CINDER through experiments on Gemma-2-2B-it (Gemma Team et al., 2024) and, across eight languages, Llama-3.1-8B-Instruct (Grattafiori et al., 2024). Notably, evaluating it exposed a blind spot in the standard protocol. EMBER's tuning score prefers edits up to a hundred times longer than the embeddings they change, which destroy the edited words rather than remove the concept from them. Because the concept questions contain the edited words, a destroyed word counts as erased knowledge, while MMLU and Alpaca coherence rarely encounter it. We therefore extend the evaluation with *collateral* questions that use the edited words in their other meanings (e.g., Prince Harry when erasing Harry Potter), one scale of edit size shared by all methods, and tests of whether the edited words survive.

Our results show that once collateral damage is charged for, every method moves from token-destroying edits to edits about as long as the token, and that embedding erasure blocks access to a concept through its words rather than removing it from the weights. Within this regime, CINDER erases concepts that EMBER misses (14 vs. 3 runs in which only one of the methods edits), and at a moderate, equal edit size its direction removes more (+0.15 on the combined score, 10 of 11 held-out concepts); this advantage vanishes at larger edits, and at each method's own operating point the two tie. Across eight languages, a single CINDER-z fit matches running EMBER separately in each language (0.23 vs. 0.25) with an eighth of the judge calls, and finds concept features for twice as many concepts as a single multilingual EMBER fit (21 vs. 10 of 38).

Finally, we analyze why the sharpened matrix finds the concept. Across 38 concepts, sharpening turns the concept from a feature that is valuable but mixed with unrelated rare tokens into one that is both pure and valuable, and finds it in more runs; global sparsity then sets its width close to where the concept's tokens actually end. The gain is not an artefact of sharpening toward the very text being scored: sharpening toward other pages about the concept keeps it, while another concept's page destroys the feature. We also find that the model's first layer reads the mean of all embeddings strongly, and that this shared direction, not the concept, explains why EMBER's edits are read more strongly than CINDER's. Concepts that live between broader topics, such as Stoicism, remain out of reach for every method.

To conclude, our work (1) diagnoses why embedding-level erasure misses quiet concepts, (2) introduces CINDER, which finds and erases them more reliably, (3) extends embedding-level erasure to many languages with a single fit, which needs an eighth of the judge calls and a fifth of the factorization work of erasing each language separately, and (4) shows that the standard evaluation of embedding erasure rewards destroying tokens, and proposes a collateral-aware alternative that needs no LLM judge inside the tuning loop.

Overall, our findings suggest that where a concept is looked for matters as much as how it is removed, and that an embedding edit must be judged by what it does to the words it edits.

2 Preliminaries and Notation

We follow the notation of Suslik et al. (2026). We consider autoregressive transformer LMs (Vaswani et al., 2017) with hidden dimension d, vocabulary $\mathcal{V}$ and input embedding matrix $E \in \mathbb{R}^{|\mathcal{V}|\times d}$, and denote by $\mathbf{e}_t \in \mathbb{R}^d$ the row of E for token $t \in \mathcal{V}$.

Concept Erasure in Language Models We follow the common setup of persistent concept erasure (Eldan and Russinovich, 2023; Li et al., 2024; Gur-Arieh et al., 2025; Suslik et al., 2026): given a model M and a target concept C (e.g., Harry Potter), we update the parameters of M so that it no longer answers questions about C better than chance, while maintaining its performance on other concepts and tasks. To this end, we assume access to a target set $\mathcal{S}_C$ of sentences about C, taken from its Wikipedia page, and a neutral set $\mathcal{S}_N$ of sentences from unrelated Wikipedia pages.

EMBER EMBER follows a localize-then-edit approach with three stages. (1) *Decomposing.* Let $\mathcal{V}^* \subset \mathcal{V}$ be the token types that appear in $\mathcal{S}_C \cup \mathcal{S}_N$, $n = |\mathcal{V}^*|$, and $A = E_{\mathcal{V}^*}^\top \in \mathbb{R}^{d\times n}$ their embeddings as columns. EMBER factorizes $A \approx ZY$, with a factor dictionary $Z \in \mathbb{R}^{d\times k}$ whose columns $\mathbf{z}_i$ are *embedding features*, and a coefficient matrix $Y \in \mathbb{R}^{k\times n}$. Sparsity is imposed on Y with a winner-takes-all constraint that keeps the 1% largest entries of each row, and the factors are optimized with alternating least squares; the authors call this Sparse MF. (2) *Identifying.* Each token is labelled concept, neutral or both, according to whether it appears in $\mathcal{S}_C$, $\mathcal{S}_N$ or both; let T_C and T_N be the concept- and neutral-labelled tokens. Each feature is scored by its mass ratio

$$\rho_i = \frac{\sum_{t\in T_C} |Y_{i,t}|}{\sum_{t\in T_N} |Y_{i,t}|}, \qquad (1)$$

and features with $\rho_i > \tau = 2$ are verified by an LLM in two stages: it first describes the feature's top tokens without seeing the concept, then decides from the description alone whether the feature is the concept, accepting at confidence ≥ 0.85. The accepted features form $\mathcal{F}_C$. (3) *Erasing.* Each concept token $t \in T_C$ with $Y_{i,t} \neq 0$ for some $i \in \mathcal{F}_C$ is edited as

$$\mathbf{e}'_t = \mathbf{e}_t - \delta\, \mathbf{c}_t, \qquad \mathbf{c}_t = \sum_{i\in\mathcal{F}_C} Y_{i,t}\, \mathbf{z}_i, \qquad (2)$$

where $\mathbf{c}_t$ is the token's *concept part* and $\delta \geq 0$ controls the erasure strength; all other embeddings remain unchanged.

Evaluating Erasure EMBER is evaluated on efficacy (accuracy on multiple-choice questions about C, which should drop to chance), specificity (accuracy on similar-domain questions and on MMLU; Hendrycks et al., 2021), coherence (AlpacaEval, judged by an LLM; Taori et al., 2023) and robustness to relearning (Deeb and Roger, 2024). Each metric is normalized by the unedited model, and their harmonic mean H selects the erasure strength $\delta \in \{0.5, 1, 2, 5, 10, 50, 100, 200\}$ on validation questions.

3 Why EMBER Misses Quiet Concepts

A natural question is how reliably EMBER's first two stages find the target concept. To study this, we reproduced EMBER on Gemma-2-2B-it, matching the published accuracies within four points (§A), and inspected each stage on its flagship concept, Harry Potter, before testing each observation on more concepts and seeds. We find five issues, summarized in Table 1.

The concept hides behind rarity Sparse MF minimizes the squared reconstruction error, so a feature is worth the energy it explains along a group of aligned tokens. Concept tokens are not low in energy (per token they carry as much as neutral tokens), but they are rare: across 38 concepts, concept-only tokens are rarer than neutral ones in 35 (median 9.4 vs. 12.9 per million words). In Gemma's embeddings, rare tokens share a large component along the background mean μ_B (Figure 2a), consistent with the degeneration of rare-token embeddings (Gao et al., 2019). The concept feature of EMBER's raw fit is therefore loud for the wrong reason: it sits at the 86th percentile of feature norms, with 42% of its norm along $\hat{\mu}_B$ against 12% for a typical feature, and it groups the concept with unrelated rare tokens. On Harry Potter (seed 42), its first 19 tokens are all Harry Potter, but further down sit Nestlé, Lego and Trump, which EMBER would edit, among unrelated names and taxonomic suffixes (§I). Remov-

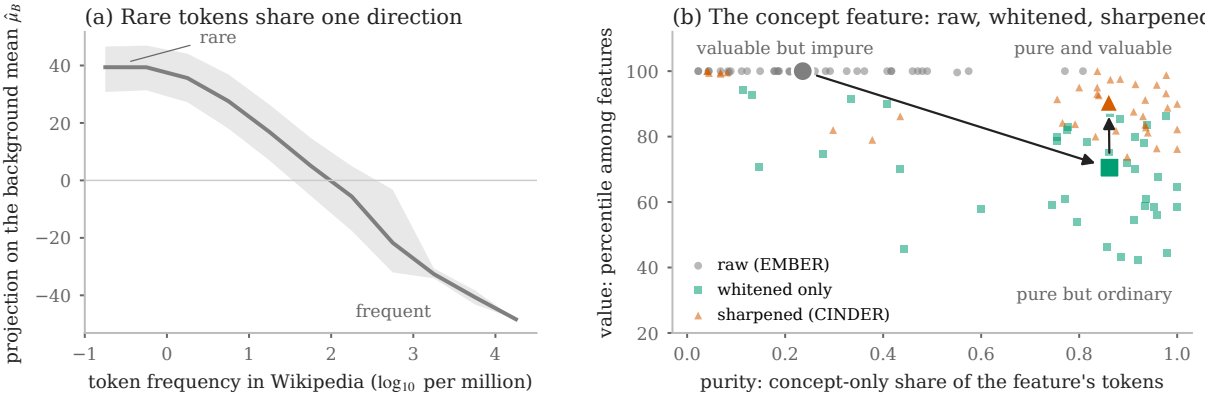


Figure 2: Why EMBER misses quiet concepts, on Gemma-2-2B-it. (a) Across the vocabulary, rarer tokens share a larger component along the background mean $\hat{\mu}_B$ (median and interquartile range per frequency bin); concept tokens are rarer than neutral ones in 35 of 38 concepts. (b) A sparse feature seeded on each concept's lexicon tokens and run to its fixed point, per concept (small markers) with medians (large markers). In the raw matrix it is valuable but drifts to other rare tokens (impure); after whitening it is pure but ordinary; after sharpening it is pure and valuable.

ing the shared component does not solve the problem: the concept's tokens then form a pure cluster, but an ordinary one, which the factorization finds less often than before (Figure 2b; §6). In this sense the concept is *quiet*.

One width does not fit all concepts EMBER keeps the top 1% of tokens in every feature, about 48 tokens. Since concepts differ in breadth, a fixed width cuts a broad concept short and pads a narrow one with unrelated tokens. To measure this, we resolve the coefficients densely with the dictionary fixed and rank all tokens by their share of the concept feature; the *natural tail* is the rank at which most of the next tokens stop being specific to the concept's page. In EMBER's matrix the 1% cut is too short for 15 of 38 concepts and too long for another 15 (§C).

The judge loses the concept in its summary EMBER's judge does not see the feature, only a one-sentence description of its top tokens, and the description is where concepts get lost. In our runs, *heroin* was described as "addictive drugs" and *World War II* as "20th-century totalitarian regimes"; both were rejected, as was Valentine's Day, each at exactly the 0.85 confidence cutoff. A rejected concept is not erased at all, since no feature is left to subtract: on 12 held-out concepts EMBER edits nothing in 19 of 60 runs, and a multilingual EMBER fit accepts a concept feature for only 10 of 38 concepts (§5).

The edit stays in English EMBER edits the rows of English tokens, and its authors leave other languages to future work. We find that a concept erased in English largely survives in other languages: on Gemma, EMBER-edited models retain 0.19–0.24 more of the concept's knowledge in German, Spanish, French and Russian than in English, and 0.31–0.41 more in Russian, whose tokens the English edit never touches (§F). The two obvious remedies both fall short. Running EMBER separately in every language multiplies the factorizations and judge calls by the number of languages, and a single EMBER fit on multilingual text accepts a concept feature for only 10 of 38 concepts on Llama-3.1-8B. In the multilingual matrix the concept is about 3 times quieter than in English, and the means of the eight languages carry 34 times the energy of the concept; the fit gives 14% of its candidate features to single languages (§5.4).

The evaluation rewards destroying tokens EMBER chooses the erasure strength δ by H on validation questions. In our reproduction, this selects the largest value, $\delta = 200$, for 6 of the 10 concepts EMBER edits. On Harry Potter, where the concept part $\mathbf{c}_t$ is about 0.6 of a token's length, the edited embedding is about 120 times longer than the original and points along $-\mathbf{c}_t$ (40–120 times across concepts): the word is replaced rather than cleaned. None of the metrics registers this. The concept questions contain the edited words, so a destroyed word counts as erased knowledge; the

Stage	Issue	CINDER's change
Decompose	The concept hides behind rarity; once separated, it is quiet	Contrastive Sharpening (§4.1)
Decompose	One width for every feature	Global Sparsity (§4.1)
Identify	The judge reads a summary that loses the concept	Two-Way Judge (§4.2)
Erase	Centred and uncentred edits differ by a direction all tokens share	Both variants (§4.3)
Scope	Edits English tokens only; one multilingual fit finds few concepts	One fit for eight languages (§4.4)
Evaluate	Destroying a word scores as erasing a concept	Collateral-aware evaluation (§5.1)

Table 1: The issues we identify in each stage of EMBER and the change CINDER makes. The erasure-stage row concerns a choice in our own design rather than an issue of EMBER.

similar-domain questions avoid them, the MMLU gate never binds in any of the 344 grid points we scored, and Alpaca coherence stays at or above 0.92. Free text tells a different story: on Harry Potter at $\delta = 200$, 12 of 20 prompts that use the edited words in other senses break (asked about Prince Harry, the model describes the singer Prince), against 0–1 of 20 at $\delta = 1$. We refer to this regime as *token destruction*.

4 CINDER

To find quiet concepts and erase them, we propose CINDER. CINDER follows EMBER's localize-then-edit approach and changes each of its three stages: (1) decomposing a contrastively sharpened embedding matrix under a global sparsity budget (§4.1), (2) identifying concept-related features with a two-way judge (§4.2), and (3) erasing them, with or without the component that all tokens share (§4.3). Finally, we extend the pipeline to many languages with a single fit (§4.4). CINDER keeps EMBER's data $\mathcal{S}_C$, $\mathcal{S}_N$ and $\mathcal{V}^*$, its Sparse MF algorithm, its ρ filter and the form of its edit (Eq. 2).

4.1 Finding Concept Features in a Sharpened Space

Sparse MF finds the directions that explain the most energy of the matrix it is given (§3). Rather than changing the factorization, we change the matrix, so that the concept carries the energy. Whitening is a standard step before sparse coding (Hyvärinen and Oja, 2000), and contrasting a target dataset with a background is a standard way to reveal what is specific to it (Abid et al., 2018); Contrastive Sharpening (CS) combines the two.

Contrastive Sharpening We first *whiten* the embeddings against ordinary text, so that directions which many tokens use no longer dominate. With the mean μ_B and covariance Σ_B of the embeddings of the token types in a large Wikipedia sample, computed once per model, each embedding is mapped to

$$\tilde{\mathbf{u}}_t = W(\mathbf{e}_t - \mu_B), \qquad W = (\Sigma_B + \varepsilon I)^{-1/2}, \tag{3}$$

so that every background direction has unit variance. We then *sharpen* the whitened embeddings toward the concept. Let $\tilde{\mathbf{u}}_1, \ldots, \tilde{\mathbf{u}}_m$ be the whitened embeddings of the token types on the concept's Wikipedia page; then

$$\mathbf{x}_t = S_\eta^{\gamma/2}\, \tilde{\mathbf{u}}_t, \qquad S_\eta = (1-\eta)\, \tfrac{1}{m} \textstyle\sum_j \tilde{\mathbf{u}}_j \tilde{\mathbf{u}}_j^\top + \eta I, \tag{4}$$

with $\eta = 0.05$ and $\gamma = 1$. Sparse MF then factorizes $X = T(A - \mu_B \mathbf{1}^\top) \approx \tilde{Z}Y$, where $T = S_\eta^{\gamma/2} W$.

Intuitively, after whitening the page's second moment along a direction measures how much more the concept uses that direction than ordinary text does. In the original coordinates, this is the ratio

$$R(\mathbf{v}) = \frac{\mathbf{v}^\top (\Sigma_C + \boldsymbol{\delta}\boldsymbol{\delta}^\top)\mathbf{v}}{\mathbf{v}^\top (\Sigma_B + \varepsilon I)\, \mathbf{v}}, \qquad \boldsymbol{\delta} = \mu_C - \mu_B, \tag{5}$$

where μ_C and Σ_C are the mean and covariance of the page's embeddings. Its stationary directions solve a generalized eigenproblem, as in Fisher's discriminant and contrastive PCA (Fisher, 1936; Abid et al., 2018), and sharpening multiplies the energy along each of them by its ratio raised to γ (Proposition 1, §B). The two steps do different jobs (§6). Whitening removes the component that rare tokens share, so the concept's tokens separate from other rare tokens into a pure cluster; but it also makes every direction equally strong, so the cluster becomes ordinary and is found less often than before. Sharpening then raises the cluster's value, so that the concept is both pure and worth a feature. When a page yields few directions that rise above sampling noise, as single English pages do on a model with a larger embedding dimension,

we sharpen only those directions; this adds no hyperparameter (§B.4).

Because T is linear and invertible and the mean cancels in a difference, the factorization maps back exactly: each token's concept part in the original space is $\mathbf{c}_t = T^{-1}\tilde{\mathbf{c}}_t$. We revisit this mapping in §4.3.

Global Sparsity EMBER's row sparsity (RS) bounds every feature separately. Global Sparsity (GS) instead bounds the whole matrix, with the same total budget:

$$\text{RS: } \|\mathbf{y}_i\|_0 \le s\,n \;\; \forall i, \qquad \text{GS: } \|Y\|_0 \le s\,k\,n. \tag{6}$$

In each alternating step, after the dense update of Y, GS scores every entry by its share of its row, $|y_{ij}|/\|\mathbf{y}_i\|_2$, and keeps the $s\,k\,n$ highest-scoring entries of the matrix, with a floor of 0.2% and a cap of 3% of the tokens per feature. Keeping the global top entries is the exact projection onto the GS constraint, as in iterative hard thresholding (Blumensath and Davies, 2009), and mirrors across features what BatchTopK does across examples (Bussmann et al., 2024). The score is scale-free on purpose: with raw magnitudes, a few high-energy generic features take the whole budget. Since the support is chosen anew in every iteration, a feature's width shapes its direction and not only the set of edited tokens (§C).

4.2 Identifying Concept Features from Both Directions

EMBER's token judge (TJ) works bottom-up: it abstracts a feature into a sentence and compares the sentence with the concept. Its failures occur when the abstraction drops the concept (§3). Inspired by query expansion in retrieval, which expands a query before matching it (Wang et al., 2023; Gao et al., 2023), the Two-Way Judge (TWJ) adds a top-down path. Once per concept, before any feature is judged, an LLM expands the concept into a lexicon L_C of 20 words specific to it rather than to its broader category. Each feature's top tokens are mapped to the words they come from in $\mathcal{S}_C$ (e.g., *ffindor* $\rightarrow$ *Gryffindor*), giving a word list W_i, and the expansion judge accepts the feature if $|W_i \cap L_C| \ge t$ (EJ@t). TWJ accepts a feature if either path does: TJ $\vee$ EJ@5. The two paths fail in different ways, since a summary can lose the concept, whereas a lexicon can miss unlisted words or match a broader feature that shares them; their union raises recall, and t controls precision. The lexicon path is deterministic and adds no LLM call per feature (§D).

4.3 Erasing Concept Features, With and Without the Mean

Because T is invertible, the sharpened factorization maps back exactly, and for fixed coefficients the least-squares dictionary does not depend on the metric in which the error is measured (Proposition 2, §E). Whitening therefore does not distort the mapped-back edit. What differs from EMBER is the centring. The sharpened fit works on embeddings centred at the background mean μ_B, so CINDER's concept part $\mathbf{c}_t$ describes how a token departs from ordinary text. EMBER factorizes raw embeddings, so its features also absorb a share of μ_B, the component that all tokens share, and its edit removes part of it as well (44% of the edit's energy on Gemma). The model's first layer reads this shared direction strongly (2.9$\times$ a random direction on Gemma, 20$\times$ on Llama), and it alone explains why EMBER's edit is read more strongly than CINDER's (§6). To compare the two on equal terms, CINDER-z keeps every decision of the sharpened fit, namely the coefficients Y, the selected features $\mathcal{F}_C$ and hence the edited tokens, and re-solves only the dictionary on the raw, uncentred embeddings:

$$Z_z = A\,Y^\top \big(YY^\top + \lambda I\big)^{-1}. \tag{7}$$

This single $k \times k$ least-squares solve needs no new fit and no further judge calls, and it adds exactly a mean term, $\mathbf{c}_t^z = \mathbf{c}_t + \beta_t\,\mu_B$ for a scalar β_t (Corollary 3). We report both variants: CINDER removes only what distinguishes the concept from ordinary text, while CINDER-z, like EMBER, also removes part of what every token shares. Whether the shared part helps to erase the concept or only makes the word atypical is an empirical question, which §5 and §7 address.

4.4 One Fit for Eight Languages

Erasing a concept in every language a model serves requires editing the tokens of every language (§3). CINDER does so with a single fit. It samples 30 concept and 30 neutral sentences in each of eight languages, the fewest that yield a full-rank matrix, and builds one matrix from all of them. In such a matrix the concept competes not only with generic topics but also with language

and script identity, which a multilingual model encodes in a few dominant directions (Chang et al., 2022; Libovický et al., 2020). CINDER therefore whitens against a background with equal amounts of text in every language, which weakens the language directions about threefold, and sharpens toward the concept's pages pooled over all languages, which gives S_η many more directions that rise above sampling noise than a single page does (§B.4). After the fit, it keeps the dictionary fixed and solves the coefficients of every token in the full multilingual sentence sets, so that tokens outside the sample can be edited as well. Features are selected with TWJ, whose lexicon covers all languages in a single LLM call, and the edit is built as in CINDER-z.

The saving is in selection. Per-language EMBER runs eight factorizations and eight judging passes per concept (160 judge calls); CINDER runs one of each (20 calls) plus one lexicon call, about a fifth of the factorization work and a third of the LLM cost of selection. The sharpening transform makes CINDER's single fit about as slow in wall-clock time as EMBER's eight, and evaluation, which dominates the total (2.5–3.5 GPU-hours per concept), costs the same for every approach, since each language must still be tested.

5 Experiments

[Skeleton. Claims and numbers come from our internal reports (blue); the experiments pass rederives each number from the result files listed in §J and fills the marked tables and figures.]

In this section, we test whether CINDER's features lead to better erasure. First, we test whether the standard protocol can tell erasing a concept from destroying its words, and introduce a collateral-aware evaluation. Second, we compare CINDER with EMBER under this evaluation on held-out concepts, and test whether the difference comes from the edit direction or from the edited tokens. Third, we test whether a single CINDER fit can replace per-language EMBER in eight languages.

5.1 Experimental Setup

Our experiments assess concept erasure on two models: Gemma-2-2B-it (Gemma Team et al., 2024) in English, and Llama-3.1-8B-Instruct (Grattafiori et al., 2024) in eight languages.

Evaluation Metrics We keep EMBER's metrics and add two that target the edited tokens:

- **Efficacy**: accuracy on multiple-choice questions about C, which a successful method drives toward chance.
- **Specificity**: accuracy on similar-domain questions and on MMLU (Hendrycks et al., 2021).
- **Coherence**: AlpacaEval scored by an LLM judge, as in EMBER (Taori et al., 2023); in the multilingual study, a perplexity ratio on held-out native text.
- **Collateral** (new): accuracy on multiple-choice questions that use an edited token in a sense outside the concept, such as *Harry* in Prince Harry or *Caesar* in Caesar salad. The items were written and reviewed by LLM agents that did not know which methods would be scored, kept only if the unedited model answers them correctly, and frozen before any edited model was evaluated (3,501 items on 1,026 tokens for the held-out concepts).
- **Word integrity** (new): whether the model can still copy a sentence containing each edited word, what it answers to "What is X?", whether a question placed after the word is still answered, and whether neutral prompts using the word keep their answer. Each test is also run on the word deleted and on a *nonce*, a pronounceable non-word that the model can repeat but does not understand, which is the target behaviour of an erased concept word.

Each metric is normalized by the unedited model. We select operating points by $H_3 = \mathrm{HM}(\text{efficacy}, \text{specificity}, \text{collateral})$, admitting only points that keep MMLU and collateral above 0.9 and 0.8 of the unedited model. Since methods differ in the length of their concept parts, equal δ does not mean equal edits; we therefore report the *edit size*

$$r = \mathrm{median}_t \frac{\|\mathbf{e}'_t - \mathbf{e}_t\|}{\|\mathbf{e}_t\|}, \tag{8}$$

where $r = 1$ means that each edit is as long as the token itself, and every method sweeps the same grid $r \in \{0.05, 0.1, 0.25, 0.5, 1, 2, 5, 10, 30, 100\}$. Because all questions are generated once, before any model is scored, evaluating a grid point requires only forward passes of the edited model, whereas EMBER's coherence term calls an LLM

judge on 150 Alpaca outputs for every point it scores. Full details are in §G.

Concepts and Data Following EMBER, $\mathcal{S}_C$ and $\mathcal{S}_N$ hold 300 Wikipedia sentences each, and each concept has 50 validation and 50 test questions. In the single-language study, we developed the evaluation on three concepts (Harry Potter, Ancient Rome and Valentine's Day) and test on 12 of EMBER's other concepts with 5 factorization seeds each (60 runs). The multilingual study uses EMBER's 18 concepts and 20 new ones, held out under a pre-registered protocol (§F).

Selection and Statistics Each method selects one operating point per run on validation questions and is scored once on test questions. Besides the primary rule (H_3 under both gates), we report the paper's rule (no collateral), a fixed equal edit size ($r = 0.5$) and an integrity-gated rule (edited words must remain copyable). Comparisons are paired by run, with the concept as the unit of analysis; we use two-sided Wilcoxon signed-rank tests with Holm correction and bootstrap 95% confidence intervals over concepts. Every analysis was written into a pre-registration before its results existed; analyses defined after earlier results were seen are marked *post hoc* below.

5.2 Erasure Methods and Baselines

Erasure methods We compare EMBER, CINDER (CS + GS + TWJ, edit mapped back), CINDER-z (edit re-solved in the original space) and, to isolate GS, CINDER-RS and CINDER-RS-z, which use EMBER's row sparsity. All methods edit only the input embeddings and use the same edit form (Eq. 2).

Multilingual approaches A1 runs EMBER separately in each language and applies all edits; A2 is a single multilingual EMBER fit; A3 is multilingual CINDER; A3z is A3 with the edit built in the original space.

Embedding editing baselines As in EMBER, we compare against replacing the edited embeddings with the vocabulary mean (Mean) and against random directions of the same length (Noise) (§G).

5.3 Results

Results are presented in Table 2, Table 4 and Figure 3.

Method, rule	r	Eff.↑	Coll.↑	Broken↓	Alp.↑
EMBER, paper	100	0.47	0.73	0.24	0.97
EMBER, ours	1	0.30	0.86	0.15	0.97
CINDER-z, paper	100	0.68	0.68	0.34	0.97
CINDER-z, ours	1	0.39	0.86	0.16	0.98

Table 2: Operating points selected by the paper's rule (no collateral) and by ours, on 12 held-out concepts (test split). r: median edit size; Broken: share of neutral prompts whose answer breaks (9 concepts where every configuration exists, one seed); Alp.: normalized Alpaca coherence (seed 42).

The standard protocol rewards token destruction Without a collateral term, selection settles on edits a hundred times the token's length (Table 2); with it, every method moves to $r \approx 1$. At the destructive points, 24–34% of neutral prompts break, and efficacy exceeds the level obtained by deleting the edited words from the questions (0.23–0.28) in 83% and 98% of runs: these edits derail the surrounding prompt as well as delete the word. Alpaca coherence registers none of this, staying at 0.97–0.98. Under the paper's rule, the original ranking reappears through destruction, with EMBER ahead of CINDER.

CINDER erases concepts that EMBER misses EMBER edits in 41 of the 60 held-out runs and CINDER in 52. In runs where only one of the two edits, CINDER does so in 14 and EMBER in 3 (exact McNemar $p = 0.013$, post hoc); EMBER's empty runs concentrate on Heroin, Nazism and Uranium.

At moderate edit size, CINDER's direction removes more At equal edit size ($r = 0.5$), every CINDER variant removes more than EMBER at the same collateral (efficacy 0.25–0.30 vs. 0.20); for CINDER the gain is $+0.15$ H_3, on 10 of 11 concepts (Holm $p = 0.012$). Crossing the two token sets with the two directions (post hoc) attributes this gain to the direction: CINDER's direction adds $+0.07$ efficacy on 10 of 11 concepts, while its token set, at matched size, adds $+0.02$. At twice the edit size ($r = 1$), the direction effect vanishes (-0.02 efficacy, $p = 0.97$; secondary analysis). One reading, consistent with §4.3, is that EMBER's uncentred edit gains from its mean component as edits grow, while CINDER's centred edit does not; the mean-edit test of §E addresses it.

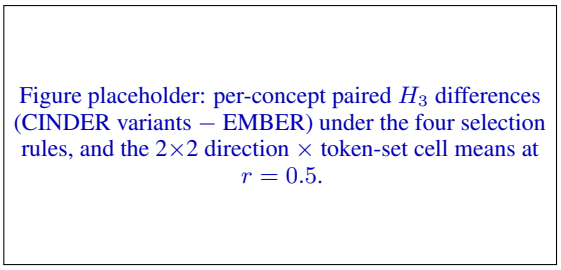


Figure 3: [**Experiments:** caption]

At their own operating points, the methods tie When each method picks its own strength, the pre-registered primary comparison shows no significant difference (+0.04, Holm $p = 0.41$): EMBER catches up with larger edits, which cost word integrity (§7). With a stricter collateral gate (0.9), all four CINDER variants beat EMBER significantly, and when runs without an edit count as failures, CINDER gains +0.12 (Holm $p = 0.10$; both post hoc, §G).

5.4 Multilingual Erasure

One CINDER fit replaces eight EMBER fits Counting concepts without an edit as failures, the single CINDER-z fit (A3z) ties per-language EMBER (A1): 0.23 vs. 0.25 strict and 0.28 vs. 0.25 ungated, 15–13 in wins and losses ($p = 0.78$; Table 4), with one factorization and one judging pass per concept instead of eight. Pooling the selections of three fit seeds, which costs three fits instead of 24, helps CINDER-z reliably (+0.08, 19–4, $p < 0.001$) and per-language EMBER not (+0.05, n.s.), and the two still tie.

A multilingual EMBER fit rarely accepts the concept Where the multilingual EMBER fit (A2) selects a concept feature, it erases as well as CINDER-z; but it selects one for only 10 of the 38 concepts, and for only 2 of the 20 held-out ones, against 21 and 10 for CINDER, consistently across three fit seeds (9–12 vs. 17–21). Counting coverage, CINDER-z beats it by +0.21 (15–4, $p = 0.011$). The loss happens at the judge more than in the fit (Table 3). The raw multilingual fit contains a lexicon-matching concept feature about as often as CINDER's (25–27 vs. 23–28 of 38 concepts per seed), but EMBER's judge accepts only 5% of its candidates with 3–4 lexicon words among their top tokens, against 30% for CINDER's and for per-language EMBER's; the raw fit also gives 14% of its candidates to single-language features, CINDER's none. Either half of CINDER's selection closes most of the gap, and together they close it fully (counterfactual coverage on the saved fits). CINDER's concept features also span more scripts (34% use two or more, against none for EMBER's), although much of their non-Latin mass is generic words.

Concepts covered (of 38)	EMBER judge	two-way judge
Raw fit	10 / 9 / 12	18 / 20 / 18
Sharpened fit (CS+GS)	16 / 16 / 16	21 / 17 / 20

Table 3: Multilingual coverage on Llama-3.1-8B by fit and judge, for fit seeds 42 / 1 / 2. The off-diagonal cells re-judge the saved fits; the diagonal cells are multilingual EMBER (A2) and CINDER (A3) as run.

	A1	A2	A3	A3z
Fits per concept	8	1	1	1
Judge calls per concept	160	20	20	20
LLM cost per concept (NIS)	.055	.007	.017	.017
Concepts edited (of 38)	26	10	21	21
Score, strict	0.25	0.11	0.14	0.23
Score, ungated	0.25	0.14	0.14	0.28

Table 4: Multilingual erasure on Llama-3.1-8B-Instruct: 38 concepts, 8 languages, test split; concepts without an edit score 0. A1: per-language EMBER; A2: multilingual EMBER; A3: multilingual CINDER; A3z: A3 with the edit built in the original space.

The mean term matters across languages Multilingual CINDER (A3), whose edit is centred, erases weakly. Re-solving its dictionary on the uncentred embeddings (A3z, defined after the development results) adds +0.17 (16 of 21 concepts, $p = 0.007$), and the gain has the same direction on the 20 pre-registered held-out concepts (+0.19, 7–3, $p = 0.064$). On Llama, the first layer reads the shared mean direction 20 times more strongly than a random one, so even a small mean term changes the edit's effect (§E); all other approaches, like EMBER, edit uncentred.

Some languages are out of reach How much is erased depends far more on the language than on the approach: most in English, about 0.6 of that in the other Latin-script languages, little in Thai and almost nothing in Hindi. Hindi and Thai words split into short pieces that also occur in everyday text, so they are never concept-only and never edited (§F); this is a limit of the concept-only rule and the tokenizer, shared by all approaches, not of the single fit.

6 Embedding-Space Analysis

A natural question is why CINDER finds concepts that EMBER misses, and why its directions remove more at moderate edit size. We expect sharpening to separate the concept from the other rare tokens and make it worth a feature, and GS to set the feature's width from the data. To test this, we analyze the features themselves, before any edit, on 38 concepts with 5 factorization seeds each (EMBER's 18 and 20 more, 15 of which were added only after the method was frozen), using judge-free measures wherever possible. Definitions and full results are in §H.

Sharpening makes the concept pure and valuable To see what each step does, we seed a single sparse feature on each concept's lexicon tokens and run it to its fixed point in five matrices: raw, raw with the mean direction $\hat{\mu}_B$ projected out, centred, whitened only, and sharpened (Figure 2b; 38 concepts, 5 seeds). In the raw matrix the feature drifts toward other rare tokens and ends up only 24% concept-only, although it is among the most valuable features of the fit. Whitening makes it pure (86%) but ordinary (70th percentile of feature energies), and sharpening keeps it pure and makes it valuable (91st percentile, higher in 37 of 38 concepts). Full fits agree: a lexicon-matching concept feature appears in 56% of raw runs, 49% of whitened-only runs and 69% of sharpened runs (better in 15 concepts, worse in 3), and it carries fewer neutral tokens (3.0 vs. 4.5). Centring alone, or removing $\hat{\mu}_B$, does not change how often the concept is found. Two observations from Harry Potter do not generalize: the raw concept feature usually already ranks near the top by ρ (median 2; Harry Potter's 12th place at seed 42 varies from 1st to 48th across seeds), and no selection-free concept direction rises to the top of the spectrum. Sharpening helps by making a clean cluster worth a feature, not by moving one direction to the top (§H.1).

The concept's neighbours become related topics In the sharpened space, the nearest neighbour of the concept feature is a related topic (magic for Harry Potter, empire for Ancient Rome, computing for Artificial intelligence) at about 74°, at the 94th percentile of an independent word-similarity measure, and the most similar 0.5% of feature pairs are semantically related (75–79th percentile, against chance for EMBER; 38 of 38 concepts). In EMBER's matrix the nearest neighbour, at about 51°, is a feature of unrelated rare tokens, and projecting out the mean direction alone opens the angle to 74°. The same direction explains the raw concept feature's large norm, which first drew our attention on Harry Potter (3rd–6th of 100 features): 97% of its excess over a typical feature lies along $\hat{\mu}_B$. Orthogonality as such is a property of the metric: each fit is near-orthogonal in its own space, and with $\hat{\mu}_B$ removed the two dictionaries are equally orthogonal in the embedding space (mean $|\cos|$ 0.08). Concepts rarely split over several features, and when they do the parts are distinct sub-topics, such as Rome and empire for Ancient Rome, which GS keeps apart (§H.2).

Widths follow the data To test whether a GS width is real, we fix the dictionary, re-solve the coefficients densely, and rank all tokens by their share of the concept feature; the natural tail is where tokens stop being specific to the concept's page (§C). The natural tail is a property of the direction: doubling the fit's sparsity budget moves it by only 3 tokens. On the sharpened matrix, EMBER's 1% cut stops before it for 25 of 38 concepts, GS widens 26 of the 28 features that are cut short, and GS widths follow the tail (Spearman 0.67 vs. 0.27 for the fixed width; mean error 30 vs. 42 tokens, $p < 0.001$), robustly across 45 definitions of the tail, though they compress its range and cut the broadest concepts short. GS changes which tokens a feature keeps, not its direction (cosine 0.97 with the row-sparse feature, within seed noise), and it helps only after CS: on the raw matrix, the width it adds is mostly neutral tokens and does not track the tail better. The width varies by about 4 tokens across seeds when the same feature is found and is unrelated to the feature's index.

The two-way judge recovers concepts lost in the summary On 19 concepts, TWJ accepts a feature that is the concept or a broader version of it in 82 of 91 runs, against 61 for EMBER's judge, and only 1 of its 110 acceptances is unrelated. Most of the added features are broader than the concept (with strict labels, the gain shrinks from 21 to 7 runs), and on 15 concepts added after the lexicon prompt was fixed, the lexicon path alone generalizes poorly, while showing the judge each token's source word does generalize (§D).

Abstract concepts stay out of reach If concepts are missed because they are quiet, EMBER should

fail most on abstract concepts that lie between louder topics. On 13 such concepts (e.g., Stoicism, Gnosticism, Gothic architecture) and 4 entity concepts, with blind labels of the top 10 features ($\kappa = 0.88$), the top-ranked feature is specific to the concept in 0 of 65 EMBER runs and 0 of 65 CS runs, against 13 and 23 of 40 for entity concepts. For six of the abstract concepts no method places a specific feature in the top 10 in any run; their top features are the parent topics (architecture, religion, painting) and discourse vocabulary. Sharpening does not recover them ($+0.04$ in top-10 presence, n.s.). We read this as a limit of embedding-level features: an abstract concept is written in a register rather than around a vocabulary of names, and the register is shared with its parent topics.

Sharpening is robust to a blurred contrast When up to half of the concept sentences are replaced by sentences from adjacent pages (The Lord of the Rings and Magic for Harry Potter), the sharpened concept feature stays in the top 10 in 80–93% of runs at a mean ρ-rank of 1.8–3.3, while EMBER's mean rank falls from 4.1 to 6.9, as much as with sentences from unrelated pages: the loss is dilution rather than lost contrast. What adjacency does cause is intrusion: neighbouring tokens (*Tolkien*, *Gandalf*) make up 14–25% of EMBER's concept feature and 5–14% of the sharpened one (§H.4).

The gain is not circular The sharpening target is the concept's own page, which contains the sampled sentences whose tokens we score. We therefore sharpened toward page sentences outside the sample, toward four other Wikipedia pages about the concept, and, as a control, toward another concept's page (18 concepts, 3 seeds, pre-registered), and scored the features with a strict label (concept-only and either a lexicon word or strongly over-represented on the page relative to Wikipedia). Both disjoint targets raise the strict purity of the concept feature as much as the page does ($+0.06$ vs. $+0.05$ over EMBER's matrix; $p = 0.003$ and 0.009), keep 66–75% of its gain in concept-vs-neutral separation, and help even on tokens that never occur in the target. Another concept's page destroys the feature (strict purity $0.57 \rightarrow 0.12$; found in 2 of 54 runs). The effect is therefore specific to the concept and does not come from scoring the sharpening text. Two weaker signals were circular or did not generalize: the coverage measure used to choose η improves only with the page target, and the ρ-rank of the concept feature does not change on average (the page lifts Harry Potter, Ancient Rome and Halloween but buries Republic of Ireland and Golf under broader co-topics). On 15 concepts outside the tuning set, every (η, γ) we tried beats the raw matrix; the default is mid-pack, and $\gamma = 2$ is slightly better (§B.3).

Features extrapolate only to a few unsampled tokens EMBER edits only tokens of the 300 sampled sentences, and the multilingual pipeline samples 30 per language, so concept tokens that occur only in unsampled sentences are missed unless the fixed dictionary extends to them. On 17 concepts with about 730 such tokens each, labelled blindly, roughly one in four extrapolated tokens is concept-specific under every rule we tried (top-10 precision 0.18–0.27, against 0.065 at random), about a third are unrelated, and re-solving the sparse codes recovers only 4% of the concept-specific tokens the sample missed. On EMBER's raw dictionary, projecting onto the concept feature mostly ranks token rarity (Spearman 0.70 with Wikipedia rarity), so a joint least-squares solve is needed; in the sharpened space the concept feature is nearly orthogonal to the rest of the dictionary (largest $|\cos|$ 0.28 vs. 0.64) and the two agree. The low precision is a property of rare tokens rather than of extrapolation: 87% of unsampled page tokens occur once on the page, and at one occurrence the in-sample support is no more precise. Extrapolation is therefore a small supplement; for the multilingual pipeline, sampling more sentences is the better lever (§H.5).

The first layer reads the shared mean, not low-variance directions CINDER's edit is read by the model's first layer about as strongly as a random direction, while EMBER's and CINDER-z's edits are read about twice as strongly (Figure 4). A natural explanation would be that whitening moves the edit into directions of low background variance, which the model ignores. The data reject it. The first layer reads low-variance directions *more* strongly than average (3.4× for the lowest 1% of the eigendirections of Σ_B), and once the background mean μ_B is projected out, EMBER's and CINDER's edits lie in the same part of the spectrum and are read alike (1.31× vs. 1.32×, 27 concepts). The whole gap is the mean term of §4.3: re-solving EMBER's own dictionary on centred embeddings lowers its read strength

to CINDER's level (1.09×), and the same holds on Llama, where the mean direction is read at 20×. Whitening thus decides which tokens an edit touches and by how much, not where in the embedding space the edit lies. The mean term is also most of what aligns an edit with the token it changes (cosine 0.50 vs. 0.30, but 0.31 vs. 0.29 without it), which suggests that part of what makes uncentred edits strong is generic rather than conceptual (§E).

7 Token-Level Analysis: What an Embedding Edit Removes

A natural concern with any embedding edit is whether it removes the concept from a word or the word from the model. Since the input embedding is where the model first reads a token, we expect an edit to control *access* to the concept through the edited words rather than the knowledge itself, and we expect stronger erasure to cost word integrity. To test this, we probe every edited word with the integrity tests of §5.1 and compare each method with two references: the word deleted, and the word replaced by a nonce.

Embedding erasure blocks access On efficacy questions that contain none of the edited tokens, every edited model answers exactly like the unedited one (efficacy 0.000 in all 18 configurations checked, including $r = 100$). The concept's knowledge thus remains in the weights and is only cut off from the edited words. At the operating points selected by our rule, efficacy lies close to the floor that deleting the edited words from the questions would produce.

Removing more costs the words At these operating points, about half of the edited words can no longer be read (0.49–0.61), and most of the concept-exclusive ones (0.65–0.81), with no significant difference between methods. Requiring the words to remain copyable (a post hoc rule) caps efficacy at 0.06–0.14; under this constraint CINDER removes about twice as much as EMBER, whereas the advantage of CINDER-z disappears, since part of it came from damage to the surrounding prompt, consistent with its extra component being a direction that every token shares (§4.3). Table 5 shows what destruction looks like in free text.

A nonce-like edit is only partly reachable An erased concept word should behave like a nonce: copyable, but meaningless. A learned edit (post hoc) that pushes each edited word toward the behaviour of its nonce keeps the words copyable and removes more than any subtraction at readable strength (2–3× EMBER's, 1.3–1.6× CINDER's), but it reaches a substantial nonce floor in only 2 of 12 concepts, and the model often can no longer name the word (§G.7). This suggests that, within a single embedding row, the concept and the word's identity share the same few directions. [**Experiments:** Figure: efficacy against word integrity (copy) along the r grid for each method, with the nonce and deletion floors.]

Prompt	$\delta = 1$ ($r \approx 0.6$)	$\delta = 200$ (selected by H)
Who is Prince **Harry**, and who is his wife?	*Prince Harry is the second son of King Charles III [. . .] His wife is Meghan Markle*	*Prince was an American singer, songwriter [. . .] Prince's wife was Mayte*
What country does the toy company **Lego** come from [. . .]?	*Lego comes from Denmark.*	*Please provide me with the name of the toy company!*
In which year was Donald **Trump** first elected President?	*[. . .] in 2016.*	*Donald Trump*

Table 5: Responses of Gemma-2-2B-it after erasing Harry Potter with EMBER (seed 42) at a moderate strength and at the strength that EMBER's selection score chooses. Edited tokens are in bold; all three belong to the Harry Potter feature. [verify r at $\delta = 1$ and that the selection chose $\delta = 200$ for this run.]

8 Related Work

Concept Erasure and Its Evaluation Most unlearning methods optimize a forget objective while retaining performance on other data (Eldan and Russinovich, 2023; Li et al., 2024; Ashuach et al., 2026), and a parallel line localizes and edits the parameters that store the concept (Gur-Arieh et al., 2025; Shafran et al., 2026; Suslik et al., 2026). Across both, erasure is often shallow: knowledge returns after relearning (Deeb and Roger, 2024), under adversarial prompting (Lynch et al., 2024), or in another format (Ye et al., 2025; Hong et al., 2025). Erasure is usually scored on efficacy, specificity and coherence, with LLM judges for free text (Zheng et al., 2023). Knowledge editing has long tested specificity on *neighborhood* prompts

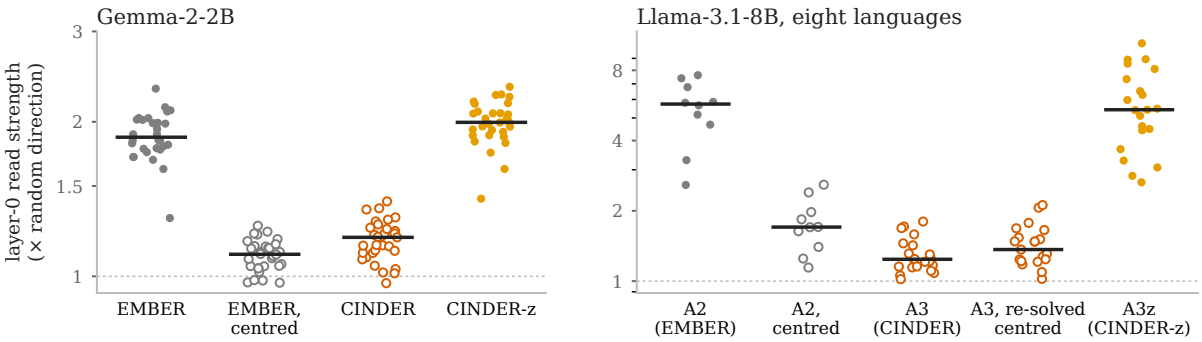


Figure 4: First-layer read strength of each edit, relative to a random direction, per concept (dots) with medians (bars). Re-solving EMBER's dictionary on centred embeddings brings its read strength down to CINDER's, and adding the mean term back (CINDER-z, A3z) raises it again, on Gemma-2-2B (left; concepts with an edit, mean over seeds) and in the multilingual Llama-3.1-8B fits (right). Filled markers: uncentred edits; open markers: centred edits.

that share surface form with the edit (Meng et al., 2022); our collateral term brings this idea to the edited tokens of embedding erasure, and our nonce reference follows the classic *wug* test of word knowledge (Berko, 1958).

Sparse Dictionaries and Their Interpretation Sparse factorizations recover interpretable parts of data (Lee and Seung, 1999; Hoyer, 2004) and of model activations (Bricken et al., 2023; Huben et al., 2024; Gao et al., 2025; Shafran et al., 2026). How the sparsity budget is allocated shapes what is found: top-k constraints fix the support of each code (Makhzani and Frey, 2014), BatchTopK shares it across a batch (Bussmann et al., 2024), and both are projections onto an ℓ_0 ball as in iterative hard thresholding (Blumensath and Davies, 2009). Features are commonly named by an LLM that explains them and scores the explanation (Bills et al., 2023; Paulo et al., 2024); since verbalized LLM confidence is poorly calibrated (Xiong et al., 2024), fixed confidence cutoffs are fragile, which motivates our second, lexical path.

Geometry of Token Embeddings Embedding spaces are anisotropic, with a few common directions dominating their variance (Ethayarajh, 2019; Mu and Viswanath, 2018), and frequency shapes their geometry, rare tokens being trained less (Gao et al., 2019). Removing or whitening the dominant directions improves semantic similarity (Mu and Viswanath, 2018; Su et al., 2021), and whitening is standard before sparse coding and ICA (Hyvärinen and Oja, 2000). Directions that separate a target from a background are classically found with Fisher's discriminant (Fisher, 1936) and contrastive PCA (Abid et al., 2018); we use the same contrast to reshape the matrix before factorizing it. Linear concept erasure methods such as INLP and LEACE (Ravfogel et al., 2020; Belrose et al., 2023) also whiten, but they remove a concept from representations at inference time, whereas we use whitening to find concept directions in the weights.

Multilingual Knowledge and Unlearning Multilingual LMs encode language identity in a few dominant directions separate from meaning (Chang et al., 2022; Libovický et al., 2020) and share much of their conceptual knowledge across languages in their middle layers (Wendler et al., 2024). Unlearning in one language often fails to carry over to others (Choi et al., 2024; Lu and Koehn, 2025). For embedding-level erasure the failure is structural, since an edit to English token rows cannot reach the tokens of another script; Suslik et al. (2026) name multilingual extension as future work, and to our knowledge no prior work erases a concept at the embedding level in more than one language. We do so with a single multilingual fit.

9 Conclusion

Embedding-level erasure can only remove the concept features it finds. We show that EMBER often misses them because concepts are quiet directions of the embedding space, overshadowed by generic topics, and because fixed feature widths and summary-based judging lose what remains. We address this with CINDER, which sharpens the embedding matrix toward the concept, lets

each feature find its width, verifies features both bottom-up and top-down, and erases them relative to ordinary text. Sharpening against a multilingual background also turns embedding-level erasure, so far limited to English, into a single fit for eight languages that matches erasing each language separately with an eighth of the judge calls. We also find that what the model's first layer reads in an edit is dominated by one direction shared by all tokens, which separates what an edit removes from a concept from what it removes from every word. Evaluating CINDER, we further show that the standard protocol rewards destroying the edited tokens, and propose a collateral-aware evaluation that charges for it. Under this evaluation, embedding erasure appears as what it is, a way to block access to a concept through its words, and within that trade-off CINDER reaches more concepts and, at moderate edit size, removes more. These findings suggest that improving embedding erasure starts with finding the concept, and that its evaluation must look at the words it edits.

Limitations

Model and language coverage Single-language results use one small model (Gemma-2-2B-it) in English, and multilingual results use one model (Llama-3.1-8B-Instruct) with a 4-bit quantized transformer and fp16 embeddings. Whether the findings transfer to other models remains to be tested.

Abstract concepts Concepts defined by a register rather than by names, such as Stoicism or Gothic architecture, yield no concept-specific embedding feature under EMBER or CINDER (§6); embedding-level erasure, with either method, does not reach them.

Embedding edits in isolation We evaluate embedding edits on their own, not ensembled with MLP-based methods as EMBER recommends, and we do not test robustness to relearning, EMBER's main strength. Whether relearning robustness also depends on the edit size ($r \approx 1$ vs. $r \approx 100$) is an open question.

Evaluation material Efficacy and collateral are multiple-choice accuracies, and collateral is lenient, since the options and context can rescue a damaged word. Collateral questions, neutral prompts and free-text labels were written and checked by LLM agents, whose blind spots may be correlated, and no human spot check was done; non-English questions are machine-translated.

Statistical power With 11–12 held-out concepts as the unit of analysis, differences below about 0.08 are hard to resolve, and several supporting analyses were defined after the primary result was known; we label them post hoc.

Sharpening target Contrastive Sharpening uses the concept's own Wikipedia page as its target, which overlaps the concept sentences. Targets disjoint from the sample keep the feature-level gains (§6), but the erasure experiments use the page target, and η and γ were chosen with it.

Ethical Considerations

CINDER is designed to remove hazardous, private or copyrighted knowledge more reliably. Like other editing techniques, it carries a dual-use risk: an effective erasure method could be used to suppress legitimate information. Our evaluation findings also carry a caution: embedding-level erasure, as commonly evaluated, can appear effective while only blocking access through specific words and damaging their other meanings, so it should not be relied on alone to remove dangerous knowledge.

AI Disclosure and Reflection

[To be completed by the team. Draft from the project records:] We used Claude Code (Anthropic's Claude models) throughout the project: to write and debug the experiment code, run and monitor cluster jobs, write and review evaluation material (collateral questions, neutral prompts, blind labels of free-text answers), analyze results, and draft and edit this report. Gemini 3.5 Flash-Lite served as the feature judge and lexicon generator (EMBER used Gemini 2.5 Flash-Lite). Every design decision, pre-registration and claim was reviewed by the authors. [Reflection: what worked, what did not, where the AI was wrong.]

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A EMBER: Reproduction Details

Sparse MF We follow EMBER's App. A.1. Given $A \in \mathbb{R}^{d\times n}$, alternating least squares updates $Y \leftarrow (Z^\top Z+\lambda I)^{-1}Z^\top A$ and $Z \leftarrow AY^\top(YY^\top+$

$\lambda I)^{-1}$ with ridge $\lambda = 10^{-4}$. After each Y update the sparsity rule is applied to the dense Y (row sparsity: the top $\lceil s\,n \rceil$ entries of each row by magnitude, $s = 0.01$), and each row of Y is rescaled to unit norm with the scale absorbed into Z. Z and Y start from i.i.d. $\mathcal{N}(0,1)$ draws; we stop when the relative error improves by less than 10^{-4} for 200 iterations and the support changes in fewer than 0.5% of its entries, or after 20,000 iterations. $k = 100$ on Gemma-2-2B-it ($d = 2304$) and $k = 200$ on Llama-3.1-8B-Instruct ($d = 4096$). Embeddings are the input embedding rows scaled by $\sqrt{d}$ for Gemma, as the model uses them.

Data EMBER's 18 concepts: Ancient Rome, Artificial Intelligence, Baseball, Cannabis, COVID-19 pandemic, Culture of Greece, Gambling, Golf, Gun, Halloween, Harry Potter, Heroin, Nazism, Pornography, Republic of Ireland, Uranium, Valentine's Day, World War II. $\mathcal{S}_C$ holds 300 sentences sampled from the concept's Wikipedia page and $\mathcal{S}_N$ 300 sentences from random English Wikipedia pages (sampling seed 42); for Harry Potter $|\mathcal{V}^*| = 4{,}780$. Questions follow EMBER: 50 validation and 50 test multiple-choice questions per concept and per similar domain, and MMLU (Hendrycks et al., 2021). The 20 additional concepts are listed in App. F.

Feature selection The 20 features with the largest ρ (Eq. 1) and $\rho > 2$ go to the judge, which uses EMBER's two prompts (its Figures 4–5) on each feature's top 20 concept-labelled tokens and accepts at confidence ≥ 0.85. Our judge is Gemini 3.5 Flash-Lite; EMBER used Gemini 2.5 Flash-Lite. Re-judging 230 candidates flipped 0.4% of decisions.

Fidelity On EMBER's protocol our reproduction reaches concept, similar-domain and MMLU accuracy of 48.4, 83.2 and 55.1, against 52.1, 84.6 and 53.6 in the paper's Table 6. Tuning with Alpaca coherence inside the δ grid, as the paper does, gives the same δ and result as our cheaper MMLU gate on Harry Potter ($H = 0.727$ vs. 0.726). In an earlier phase we found that a 4-token answer budget in the original multiple-choice scorer counted many correct answers as wrong at large edit strengths, inflating EMBER's efficacy. All erasure numbers here use the repaired scorer, except the cross-lingual leakage of the first study (App. F), which we report with the original scorer and with unparseable answers dropped.

B Contrastive Sharpening: Details

B.1 Construction

Background The background is the set of token types in a sample of English Wikipedia, each type counted once (107,472 types for Gemma), so that a direction is loud in the background because many *different* tokens use it, not because a few frequent ones do. We compute its mean μ_B and covariance $\Sigma_B = Q\Lambda Q^\top$ once per model. The ridge $\varepsilon = 10^{-4}\cdot\mathrm{mean}(\Lambda)$ only damps the smallest eigenvalues.

Target The sharpening target is the set of token types of the concept's English Wikipedia page ($m = 3{,}124$ for Harry Potter), whitened with Eq. 3. $\eta = 0.05$ and $\gamma = 1$ were chosen by a judge-free sweep on Harry Potter, Ancient Rome and Valentine's Day: each of five values of η was ranked by concept-vs-neutral AUC, concept-only purity of the top feature, concept-token coverage and $\rho_{\max}$, and the lowest mean rank won. $\eta = 1$ is whitening only; γ sets how strongly enriched directions are stretched.

Exact inverse In column form, $\mathbf{x}_t = T(\mathbf{e}_t - \mu_B)$ with $T = S_\eta^{\gamma/2} W$, which is invertible because $S_\eta \succeq \eta I$ and $W \succ 0$. If sparse MF gives $\mathbf{x}_t \approx \sum_i Y_{i,t}\tilde{\mathbf{z}}_i$, then $\mathbf{e}_t - \mu_B \approx \sum_i Y_{i,t}\, T^{-1}\tilde{\mathbf{z}}_i$: the same coefficients reconstruct the original embedding with the mapped-back dictionary $\mathbf{z}_i = T^{-1}\tilde{\mathbf{z}}_i = W^{-1}S_\eta^{-\gamma/2}\tilde{\mathbf{z}}_i$, and the mean cancels in any difference. Every erasure run checks numerically that the transformed embedding rows equal the factorized matrix columns before editing.

B.2 Why sharpening finds the concept

Let $M = \frac{1}{m}\sum_j \tilde{\mathbf{u}}_j\tilde{\mathbf{u}}_j^\top = P\,\mathrm{diag}(\lambda_j)\,P^\top$ be the whitened second moment of the target, so that $S_\eta = P\,\mathrm{diag}(g_j)\,P^\top$ with $g_j = (1-\eta)\lambda_j + \eta$.

Proposition 1. *(i) The eigenvalues of M are the generalized eigenvalues of the pair $(\Sigma_C + \boldsymbol{\delta}\boldsymbol{\delta}^\top, \Sigma_B + \varepsilon I)$, i.e. the stationary values of the ratio R of Eq. 5, attained at $\mathbf{v}_j = W\mathbf{p}_j$. (ii) For any set of tokens with whitened second moment G, the second moment of the sharpened tokens along $\mathbf{p}_j$ is $g_j^\gamma\,\mathbf{p}_j^\top G\,\mathbf{p}_j$. In particular, background tokens, for which $G = I$ up to ε, carry the same energy along every direction after whitening and g_j^γ after sharpening, and the target's own energy along $\mathbf{p}_j$ becomes $g_j^\gamma\lambda_j \approx (1-\eta)^\gamma\lambda_j^{1+\gamma}$ for $\lambda_j \gg \eta$.*

Proof sketch. (i) $\mathbf{p}^\top M\mathbf{p} = \mathbf{p}^\top W(\Sigma_C + \boldsymbol{\delta}\boldsymbol{\delta}^\top)W\mathbf{p}$, since the target's second moment around μ_B is $\Sigma_C + \boldsymbol{\delta}\boldsymbol{\delta}^\top$; with $\mathbf{v} = W\mathbf{p}$ and $\mathbf{p}^\top\mathbf{p} = \mathbf{v}^\top W^{-2}\mathbf{v} = \mathbf{v}^\top(\Sigma_B+\varepsilon I)\mathbf{v}$, the Rayleigh quotient of M equals $R(\mathbf{v})$. (ii) $S_\eta^{\gamma/2}\mathbf{p}_j = g_j^{\gamma/2}\mathbf{p}_j$, so $\mathbf{p}_j^\top S_\eta^{\gamma/2} G S_\eta^{\gamma/2}\mathbf{p}_j = g_j^\gamma\, \mathbf{p}_j^\top G\mathbf{p}_j$. □

Since the error that a sparse-MF feature removes is the energy of its support along it, the proposition says which directions sparse MF will find profitable. In the raw matrix they include the mean direction that rare tokens share, so a concept's rare tokens are grouped with other rare tokens. After whitening, all background directions are equally profitable, so the concept's tokens form a pure cluster of ordinary value; whitening alone finds a concept feature less often than the raw matrix (on Harry Potter it merges the concept with magic). After sharpening, they are ordered by the concept-to-background ratio, raised to the power $1+\gamma$ for the concept's own tokens. Two further facts make this precise. First, EMBER's alternating fit depends on A only through the token Gram matrix $K = A^\top A$: with $G_Y = YY^\top + \lambda I$, the dictionary update gives $Z^\top Z = G_Y^{-1} Y K Y^\top G_Y^{-1}$ and $Z^\top A = G_Y^{-1} Y K$, so the dense solve, the masks and the error are functions of K (verified numerically: identical supports for A and QA, Q orthogonal). Under CS with $\gamma = 1$, K becomes a *page kernel*, $K(s,t) = (1-\eta)\frac{1}{m}\sum_{j\in\text{page}} k(s,j)\,k(t,j) + \eta\, k(s,t)$, with k the background-Mahalanobis inner product: two tokens are similar when they resemble the same tokens of the concept page. Second, with the other features fixed, the best feature on a support S of fixed size lowers the error by $\sigma_1(R_S)^2 = \left(\sum_{t\in S}\|\mathbf{r}_t\|^2\right)\kappa(S)$, the support's energy times its coherence $\kappa(S) \in [1/|S|, 1]$. A concept can therefore fail in two ways: it is *impure* if other tokens project more strongly along its best direction (the raw case, where rare tokens share $\hat{\mu}_B$), or *unprofitable* if its energy times coherence is below what other directions offer (the whitened-only case). Across concepts the largest gain is $g_1 = 15.7$ (median; range 10.9–29.3), and 870 of 2,304 directions have $g_j > 1$.

Decomposition Medians over 38 concepts × 5 seeds, row sparsity:

	raw	$\perp \hat{\mu}_B$	centred	whit.	CS
Concept found (% runs)	56	61	59	49	69
Its neutral tokens	4.5	4.5	4.5	4.0	3.0
Its $\|\mathbf{z}\|^2$ percentile	86	66	41	54	58
Seeded: purity	.24	.27	.04	.86	.86
Seeded: value pct.	100	97	100	70	91
SVD rank matching	4	22	4	66	49

The seeded feature starts on the concept's lexicon tokens and alternates between the top singular vector of its support and the top tokens along it until the support repeats.

B.3 Is the gain circular?

Design 18 EMBER concepts, seeds 42–44, row sparsity (global sparsity as a secondary analysis), everything fixed except the sharpening target, preregistered before any new fit. Targets: the page (default); *disjoint*, page sentences that are not near-duplicates of any sampled sentence; *related*, four other Wikipedia pages about the concept, taken by rule from the page's "Main", "Further" and "See also" links and size-matched; a random page subset as large as *disjoint*; the leak-free control (page minus every token of $\mathcal{V}^*$); and *wrong*, the page of an unrelated concept. A token is *strict* if it is concept-only and either a lexicon word or occurs at least twice in $\mathcal{S}_C$ and is at least ten times more frequent there than in Wikipedia.

Results Strict purity of the best concept feature (raw .568): page +.052 ($p = .088$), disjoint +.060 (.003; Holm .034), related +.062 (.009), page subset +.046, leak-free −.064 (n.s.), wrong −.445 ($p < 10^{-4}$). Separability of strict and neutral tokens (AUC, raw .576): page +.055 (17 of 18 concepts), related +.041 (17 of 18), disjoint +.037 (15 of 18); on tokens absent from the target, related still gives +.034 ($p = .0003$). With the wrong page the concept feature is found in 2 of 54 runs (raw 39). The coverage measure used to choose η rises only with the page target (+.033) and falls with the disjoint and related ones, so it was the circular quantity. The target also steers which side of the concept is amplified: for Harry Potter, the page keeps the proper-name feature, while the disjoint and related targets yield a "magic" feature, both ranked first. Under global sparsity the purity gain disappears for every target, page included, while the separability gains remain.

Sensitivity to η and γ On the 15 concepts outside the tuning set (page target, 3 seeds), every setting of $\eta \in \{0.05, 0.2, 0.5\}$ and $\gamma \in \{0.5, 1, 2\}$

beats the raw matrix on strict purity, separability and ρ-rank; the default ranks 6th of 9, $\gamma = 2$ is slightly better ($+.024$ purity), and $\gamma = 0.5$ is worst on every measure.

B.4 Models with a larger embedding dimension

The question S_η is estimated from the m token types of one page. On Llama-3.1-8B ($d = 4096$), 15 of the 18 English pages have $m < d$ (median $m/d = 0.77$; Gemma 1.29), so $(1-\eta)M$ has a null space that sharpening shrinks to $\eta^{\gamma/2}$.

Rank is not the problem The fitted features put under 0.5% of their mapped-back norm into these floor directions; the three Llama pages with $m > d$ fail like the others; and Gemma pages cut below d, or Gemma's own pages with $m < d$, work as before.

Noise is Only about 12 eigenvalues of Llama's whitened page moment exceed the largest eigenvalue of the same moment computed from m random background types (Gemma: 74), and in the noise-dominated bulk each page token's own outer product supplies about half of its sharpened energy (predicted $d/(d+m)$; measured median 0.54, correlation 0.99 across 36 model–concept pairs). Since almost every concept-only token of $\mathcal{V}^*$ is on the page, sharpening then amplifies page membership rather than the concept: on Llama, 92% of features pass $\rho \geq 2$ (EMBER 32%), their words are close to incoherent, and the concept feature falls from a ρ-rank of about 10 to about 38 ($p = 10^{-4}$, 18 concepts $\times$ 3 seeds). No $\eta \in [0.05, 0.5]$ or $\gamma \in \{0.5, 1\}$ changes this, since the echo sits in the bulk, not on the floor.

Rule Sharpen only the resolvable directions, $S = I + \sum_{j:\lambda_j > \hat{\lambda}_{\text{edge}}} \left((1-\eta)\lambda_j + \eta - 1\right) \mathbf{p}_j \mathbf{p}_j^\top$, with $\hat{\lambda}_{\text{edge}}$ the largest eigenvalue of the random-type moment (the Marchenko–Pastur edge $(1 + \sqrt{d/m})^2$ is a close closed form). On Llama this gives a concept-feature rank of about 5 against 10 for EMBER ($p = 0.015$) with EMBER's coherence, though no better than whitening alone; on Gemma it keeps 47–98 directions and changes nothing. Pooling the concept's text over eight languages raises the number of resolvable directions to about 85, which, with the same rule, also works (rank 4.5).

Algorithm 1 Sparse MF with Global Sparsity (one iteration)

Require: $A \in \mathbb{R}^{d \times n}$, $Z \in \mathbb{R}^{d \times k}$, budget $K = \lceil s\,n \rceil k$, floor $m_{\min}$, cap $m_{\max}$

1: $Y^{\mathrm{d}} \leftarrow (Z^\top Z + \lambda I)^{-1} Z^\top A$ ▷ dense update
2: $s_{ij} \leftarrow |Y^{\mathrm{d}}_{ij}| \,/\, \|Y^{\mathrm{d}}_{i,:}\|_2$ ▷ scale-free share
3: $\mathcal{S} \leftarrow$ the top $m_{\min}$ entries of every row by s_{ij}
4: add the remaining entries in decreasing s_{ij}, skipping rows with $m_{\max}$ entries, until $|\mathcal{S}| = K$
5: $Y \leftarrow Y^{\mathrm{d}} \odot \mathbb{1}_{\mathcal{S}}$; rescale rows of Y to unit norm, absorb scales in Z
6: $Z \leftarrow AY^\top (YY^\top + \lambda I)^{-1}$

C Global Sparsity: Details

Algorithm Algorithm 1 replaces EMBER's per-row top-k step; everything else is unchanged. The budget equals EMBER's total (1% of all entries); the floor is 0.2% and the cap 3% of the tokens per feature ($m_{\min} = 10$, $m_{\max} = 144$ on Harry Potter). The cap exists for generic features; no concept feature reached it in 190 runs over 38 concepts (widest: Culture of Greece, 120 of 156 tokens).

Variants All five rules below share the matrix, the budget and the fitting loop.

Rule	Support	Role
RS-1%	top 1% of each row	EMBER
RS-2%	top 2% of each row	more tokens, no reallocation
GS-norm	global top-K of $\lvert y_{ij}\rvert / \lVert \mathbf{y}_i \rVert$	CINDER
GS-gain	global top-K of $\lvert y_{ij}\rvert \, \lVert \mathbf{z}_i \rVert$	favours loud features
GS-entropy	width $\propto$ effective support $e^{H(\mathbf{y}_i)}$	ablation

An ℓ_1-penalized variant (the Lasso, Tibshirani 1996), tuned to the same number of nonzeros, never produced a clean Ancient Rome feature in 5 seeds and was dropped.

Results On the sharpened matrix, 18 concepts $\times$ 5 seeds (cap 5% in this sweep):

	RS-1%	RS-2%	GS-norm	GS-gain	GS-ent.
Width	43–52	86–104	15–113	9–143	25–121
Seed MAD	0	0	4.1	8.1	7.9
Clean runs (/90)	75	31	76	55	61
Neutral share	5.8%	8.2%	5.6%	8.0%	8.2%
Seed Jaccard	.719	.726	.735	.717	.733

MAD: mean absolute deviation of a concept's width across seeds. A run is clean when its concept feature has at most five neutral tokens. Only GS-norm adapts the width per concept while staying as clean and stable as EMBER's rule. This

sweep used a 5% cap; with the current 3% cap, concept widths range over 21–114 tokens (concept means).

Width, tails and breadth We use the saved fits of 38 concepts × 5 seeds (EMBER; CS with row sparsity; CS+GS; raw GS; CS with 2% row sparsity), 950 fits in all. *Dense re-solve.* With the saved dictionary fixed, $Y' = (Z^\top Z + \lambda I)^{-1} Z^\top A$ equals Y on the support in every fit (maximum error $2.2 \cdot 10^{-6}$), and every GS support is exactly the set of tokens whose share exceeds a single threshold per fit. *Natural tail.* Along the dense ranking of the concept feature (identified by lexicon hits), a token is *page-enriched* if it occurs in $\mathcal{S}_C$ and is at least 20 times more frequent there than in Wikipedia; the natural tail T is the first rank at which fewer than half of the next 20 tokens are page-enriched, fixed before any width statistic was computed. *Results.* On the sharpened matrix the median T is 69 tokens against a row-sparse width of 47; the 1% cut is short for 25, long for 8 and about right for 5 of 38 concepts (raw matrix: 15, 15, 8). Doubling the budget moves T by +3 tokens (rank correlation 0.96). GS widens 26 of the 28 features cut short; its width correlates with T at 0.67 [0.42, 0.84] against 0.27 for the row-sparse width, with mean error 30 vs. 42 tokens ($p = 0.0009$). This holds for all 45 non-degenerate definitions of the tail in a grid of 72, but GS compresses the range (slope 0.30): it overshoots narrow concepts by 24 tokens and undershoots broad ones by 44 (e.g., Buddhism 100 vs. 182). *Stability.* The width's seed deviation is 4 tokens when the same feature is identified (7.7 overall), and the width is unrelated to the feature's index ($\rho = -0.004$). *Breadth.* GS width does not correlate with counts of concept tokens (0.05); that count ranks concepts exactly like $|\mathcal{V}^*|$, since $\mathcal{S}_N$ is shared. *Direction.* The GS and row-sparse concept features have cosine 0.97 (within seed noise); raw and sharpened features, mapped back, only 0.75. *GS needs CS.* On the raw matrix, the width that GS adds beyond the row-sparse width is 16% neutral (against 7% for its first 47 tokens), and raw GS does not track T better than row sparsity ($p = 0.39$). Under GS the concept feature ranks first by ρ less often than under CS with row sparsity (18% vs. 38% of runs), although its wider support holds more concept tokens (54 vs. 41.5).

D Two-Way Judge: Details

Lexicon Once per concept, before any feature is judged, the LLM receives:

> *Given the concept '{concept}', list the {n} words most strongly associated with it. Rules: Start with the concept's own name and its basic word forms when they are single words [...]. Then the entities, people, places, objects, actions and terms that an ordinary person with general knowledge would immediately connect to this concept. Prefer well-known words over expert or niche terminology. Every word must be specific to this concept: avoid words that fit its broader category just as well (e.g. for 'World War II' not 'war', 'battle' or 'soldier'). Single words, lowercase, no duplicates or mere plural variants. Return ONLY a JSON list of {n} strings.*

with $n = 20$. The prompt's examples name four concepts (Culture of Greece, Ancient Rome, Artificial Intelligence, World War II), which we exclude when evaluating the judge. In the multilingual pipeline the same call returns 20 words in each of the eight languages.

Feature words For each of a feature's top tokens we take the word in $\mathcal{S}_C$ that most often produces it (*ffindor* → *gryffindor*), lowercase and deduplicate the words, and keep the first 20; EJ@t accepts when at least t of them are lexicon words.

Variants

Rule	Accepts when
TJ	EMBER's two-stage judge accepts
WJ	TJ, shown each token's source words
WJ+share	WJ, and the concept share is ≥ 0.5
EJ@t	$\geq t$ lexicon words among the feature's words
TWJ	TJ ∨ EJ@5
TWJ-R	WJ+share ∨ EJ@4

Labelled evaluation Every rule scored every candidate feature of the CS+GS fits (23 concepts × 5 seeds, 7,105 candidates). Every feature accepted by some rule was labelled, without seeing the rules' decisions, as P (specific to the concept), B (broader or adjacent, e.g. nuclear physics for Uranium) or N (unrelated). Excluding the four prompt-example concepts leaves 19 concepts and 95 runs; a run is hit when an accepted feature is positive.

	P only			P + B		
Rule	Prec.	Rec.	Hit/72	Prec.	Rec.	Hit/91
TJ	.85	.69	60	1.00	.41	61
WJ	.56	.81	61	.77	.55	66
EJ@5	.74	.64	64	.99	.43	81
TWJ	.71	.77	67	.99	.54	82
TWJ-R	.61	.90	71	.91	.68	84

Recall counts only positives that some rule accepted. The threshold $t = 5$ was chosen on this set. On the 15 concepts added after the prompt was fixed, EJ@5 does not generalize (precision 0.46, recall 0.26) and TWJ falls to 0.53/0.56, whereas TJ $\vee$ WJ+share keeps precision 0.71 and recall 0.89 (49 of 75 runs hit, against 33 for TJ). In erasure the two judges perform the same. [**Experiments:** Final judge comparison table on the held-out concepts, and a per-concept list of TJ's wrong rejections.]

Cost TJ makes two LLM calls per candidate feature (up to 20 per run). The lexicon path makes one call per concept, shared by all seeds and fits, and its matching is deterministic, so its decisions are reproducible and free to recompute after any change to the factorization.

E Centred and Uncentred Edits

Read strength The first thing the model does with an embedding is normalize it and project it into queries, keys and values. For a unit direction $\mathbf{u}$ we define its read strength as $\left\|[W_Q; W_K; W_V]\,(\mathbf{g} \odot \mathbf{u})\right\|$, with the layer-0 attention projections stacked and $\mathbf{g}$ the layer-0 RMSNorm scale, relative to the mean over 500 random unit directions; for Llama we also stack the MLP's gate and up projections, as the multilingual study did. For a method we take the median over its edited tokens' directions $\mathbf{c}_t/\|\mathbf{c}_t\|$.

The dictionary step does not depend on the metric Write $A_c = A - \mu_B \mathbf{1}^\top$ for the centred embeddings. The sharpened fit alternates $\tilde{Z} \leftarrow \tilde{X}Y^\top(YY^\top + \lambda I)^{-1}$ on $\tilde{X} = TA_c$ with a masked coefficient update, and CINDER maps the dictionary back as $T^{-1}\tilde{Z}$.

Proposition 2. *For invertible T, $B \in \mathbb{R}^{d\times n}$, $Y \in \mathbb{R}^{k\times n}$ and $\lambda \geq 0$ with $YY^\top + \lambda I \succ 0$,*

$$\arg\min_Z \|T(B - ZY)\|_F^2 + \lambda\|TZ\|_F^2 = BY^\top(YY^\top + \lambda I)^{-1},$$

independently of T.

Proof. $Z \mapsto \tilde{Z} = TZ$ is a bijection, and in $\tilde{Z}$ the objective $\|TB - \tilde{Z}Y\|_F^2 + \lambda\|\tilde{Z}\|_F^2$ is strictly convex with minimizer $\tilde{Z}^* = TBY^\top(YY^\top + \lambda I)^{-1}$; hence $Z^* = T^{-1}\tilde{Z}^* = BY^\top(YY^\top + \lambda I)^{-1}$. □

At convergence of the fit, the mapped-back dictionary is therefore the Euclidean least-squares dictionary of the *centred* embeddings, $Z_c = A_cY^\top(YY^\top + \lambda I)^{-1}$, for the affine model $\mathbf{e}_t \approx \mu_B + Z\mathbf{y}_t$ (on Gemma, $\|T^{-1}\tilde{Z} - Z_c\|/\|Z_c\| \leq 0.008$ in all 137 runs). The coefficient step is where the metric matters: Y is an M-orthogonal projection with $M = T^\top T$, so whitening chooses which tokens are grouped and how they are weighted.

Proposition 3. *CINDER-z's dictionary (Eq. 7) is $Z_z = Z_c + \mu_B\mathbf{b}^\top$ with $\mathbf{b} = (YY^\top + \lambda I)^{-1}Y\mathbf{1}$, so its edit is $\mathbf{c}_t^z = \mathbf{c}_t + \beta_t\mu_B$ with $\beta_t = \sum_{i\in\mathcal{F}_C} b_iY_{i,t}$.*

Proof. Substitute $A = A_c + \mu_B\mathbf{1}^\top$ in Eq. 7. □

In the data, $\mathbf{c}_t^z - \mathbf{c}_t$ has at least 0.9995 of its energy along μ_B in every run (Gemma 137, Llama 21). For a feature of width w with same-sign coefficients of size about $1/\sqrt{w}$, $\beta_t \approx 1$ whatever w, so at $\delta = 1$ an uncentred edit subtracts roughly one full μ_B (observed median β: EMBER 0.66, CINDER-z 0.56, CINDER 0.05; the edited tokens' own coordinate along μ_B is 1.06). The affine model also reconstructs raw embeddings better than the uncentred one (relative error 0.917 vs. 0.932, 137 of 137 runs).

Where the read-strength gap comes from Concept means on Gemma-2-2B; $\perp\ \mu$: with $\hat{\mu}_B$ projected out of the edit.

Edit	Read	$\hat{\mu}_B$ share	Read $\perp\ \mu$	$\cos(\mathbf{c}_t, \mathbf{e}_t)$
EMBER	1.86	0.44	1.31	0.50
EMBER, centred Z	1.09	0.07	1.31	0.35
CINDER	1.18	0.05	1.32	0.30
CINDER-z	1.99	0.48	1.32	0.50

EMBER vs. CINDER: +0.68 read (27 of 27 concepts), but −0.01 once $\hat{\mu}_B$ is removed ($p = 0.44$), with the same Rayleigh ratio of the background covariance along the edit (4.3 vs. 4.4). Across all edits, read strength follows the share along $\hat{\mu}_B$ (Spearman 0.75) more closely than the background variance along the edit (0.63). The read strength of the eigendirections of Σ_B is U-shaped in their variance (top decile 1.08, middle deciles 0.81–0.88, bottom decile 1.59, lowest 1% 3.4); strongly read directions concentrate on a few coordinates with a large first-layer normalization

weight. On Llama, A3z exceeds A3 by +4.48 in read strength, of which +4.34 is the mean term; $\hat{\mu}_B$ is read at 20.4×, so a 7% share suffices. [**Analysis pending:** WP6 follow-ups not yet run: a pure mean edit $\mathbf{e}_t - \beta_t \mu_B$ on CINDER's tokens, to test whether the mean term erases the concept or only makes the word atypical; read strength with the normalization weight set to one; forward-pass sensitivity instead of the linear proxy.]

F Multilingual Erasure: Details

English erasure leaks In a first study we erased EMBER's 18 concepts from Gemma-2-2B-it with EMBER, SNMF (Shafran et al., 2026) and their combination, tuned every edit in English, and asked the held-out questions in English, German, Spanish, French and Russian. Retention, the share of the concept knowledge left (1 = as before, 0 = chance), was on average 0.24 higher in the four other languages than in English under EMBER (95% CI [0.15, 0.32]; 0.19 [0.12, 0.27] with unparseable answers dropped), and 0.41 (0.31) higher in Russian, where the English edit touches none of the concept's tokens. These numbers use the original scorer of App. A. SNMF, which edits MLP weights, erased all five languages about equally (+0.01), and adding EMBER to it brought the gap back (+0.16).

Languages, model and data Llama-3.1-8B-Instruct in its eight supported languages: English, German, French, Italian, Portuguese, Hindi, Spanish and Thai. The transformer runs in 4-bit NF4 and the edited embeddings in fp16, identically for every approach. Concept sentences come from the concept's native Wikipedia page (via inter-language links), up to 300 per language, topped up with NLLB translations of English sentences for short pages; neutral sentences come from native Wikipedia pages. Shared fits use 30 concept and 30 neutral sentences per language. The 20 held-out concepts are French Revolution, Tuberculosis, Cricket, Buddhism, Plutonium, Judaism, Karate, Existentialism, Prime number, The Matrix, Louvre, American Revolutionary War, Robot, Sigmund Freud, Giraffe, Surfing, Flush toilet, Cloud, Doctor of Philosophy and Tourism. EMBER's English questions are used as published; German, French and Spanish come from existing translations; Italian, Portuguese, Hindi and Thai are machine-translated (NLLB and MADLAD candidates ranked by LaBSE, with script and number checks). Questions for the new concepts were generated from the English page in EMBER's format, checked by a blind verifier and by automatic checks, and translated the same way.

Metrics and gates Per language L: efficacy (chance-corrected concept accuracy driven to chance), specificity (harmonic mean of Global-MMLU and similar-domain retention), fluency ($\min(1, \text{ppl}_{\text{orig}}/\text{ppl}_{\text{edit}})$ on 100 held-out native sentences), and H_L, their harmonic mean. A concept's score is the mean H_L over eligible languages (unedited accuracy ≥ 0.4). $\delta \in \{0.5, \ldots, 200\}$ is chosen on validation, one value for all languages, and fails if MMLU or fluency drops below 0.9 of the unedited model in any language. We report the *strict* score (0 if a gate fails on test) and the *ungated* one.

Approaches

	Fit	Selection	Edit
A1	8 per-language EMBER fits	TJ per language	each language's; overlaps resolved by page length
A2	1 fit on the 8-language sample, Y extrapolated	TJ	EMBER's
A3	A2 with CS (8-language background) and GS	TWJ	mapped back
A3z	as A3	as A3	Z re-solved on raw E

Results [**Experiments:** Development table (18 concepts) and the pre-registered hypotheses H1–H3 on the 20 held-out concepts, with paired differences, CIs, win–loss counts and Wilcoxon p (source: `results/ml/PREREG_RESULTS.md`); the seed-coverage table; the seed-union follow-up (pooling selections over 3 fit seeds: +0.08 for A3z, 19–4, $p < 0.001$; pooled A1 and A3z tie at 0.30 vs. 0.31).]

Language identity in the single fit Each token of the multilingual $\mathcal{V}^*$ is assigned to a language when at least 80% of its occurrences, normalized by each language's token count, come from that language. In the raw sample matrix, the language means carry 1.6% of the centred energy against 0.047% for the concept-vs-neutral mean (median ratio 34×), the concept is 2.9× quieter than in an English-only matrix (38 of 38 concepts), and the top principal directions are explained by language

($\eta^2 = 0.25$) rather than by the concept (0.004). Whitening against the eight-language background halves the language share (1.6% → 0.8%) and cuts the enrichment of the top contrastive directions from 8.2 (English background) to 2.6, but those directions remain single-language clusters. 8.6% of the raw fit's features and 13.7% of its judged candidates are dominated by one language (mostly German and English function words), against 4.9% and 0% for the sharpened fit. The raw fit nevertheless contains a lexicon-matching concept feature in 25–27 of 38 concepts per seed (sharpened: 23–28); the token judge accepts such features far less often when they come from the raw fit (5% vs. 30% at 3–4 lexicon words), e.g. describing Cannabis's *marijuana, cannabis, alcohol, tobacco, THC* as "psychoactive substances". Blind labels of the features accepted only by the lexicon path are specific or broader, never unrelated, for both fits.

Selection cost Per concept (fit seed 42), per-language EMBER runs 8 factorizations over about 30,600 rows with 160 judge calls (0.055 NIS); the single fits run one factorization over about 6,500 sampled rows plus a coefficient solve for the full set, with 20 judge calls (EMBER: 0.007 NIS) or 20 calls and one lexicon call (CINDER: 0.017 NIS, 0.007 once the lexicon is cached across seeds). Factorization work (rows × iterations) drops about fivefold, but CINDER's wall-clock fit time (40 s on a GPU) matches EMBER's eight fits (44 s) because of the sharpening transform. Evaluation takes 2.5–3.5 GPU-hours per concept and approach, about 250 times the fit stage, and is the same for every approach.

Why Hindi and Thai resist EMBER edits only concept-only tokens, which never occur in the neutral sentences. Llama splits Hindi and Thai words into short pieces that also occur in everyday text, so almost none qualify: on Baseball, 32% of English question tokens are concept-only, against 13% in German, 6% in Thai and 5% in Hindi, and the share of questions containing an edited token falls from 0.94 (English) to 0.00 (Hindi). The same mechanism lets an edit made for one language act in every language that shares its pieces: Golf's per-language fit found its only feature in Thai, whose tokens cover 76% of the English questions.

Robustness Counts of concepts with an edit are stable across three fit seeds (A2: 9–12; A3: 17–21 of 38), but which concepts are edited is not, and the edited token sets of one concept overlap by a Jaccard index of only ≈0.2 across seeds, for every approach. In 6 runs, across all approaches, a δ that passed both gates on validation failed one on test.

G Collateral-Aware Evaluation: Details

G.1 Collateral Questions

For every edited token with a common meaning outside the concept (e.g., *Harry*, *Caesar*, *club*), LLM agents (Claude) wrote four-option multiple-choice questions that use the token in that meaning, and other agents reviewed them, both without knowing which methods would be scored. Items that the unedited model answers incorrectly under either option order were dropped, and each split was frozen with a hash before any edited model was scored. The held-out set has 3,501 items on 1,026 tokens. Collateral accuracy is normalized against the unedited model per token, like EMBER's metrics.

G.2 Scores, Gates and Selection Rules

H_3 is the harmonic mean of efficacy, specificity and collateral; H_4 adds Alpaca coherence. A point is admissible only if MMLU stays at or above 0.9 and validation collateral at or above 0.8 of the unedited model. Each rule selects one grid point per run on validation and scores it once on test:

Rule	Selects	Role
Collateral	max H_3, both gates	primary
Paper	max HM(efficacy, specificity), MMLU gate	the old protocol
Fixed	the point $r = 0.5$	equal edit size
Integrity	collateral rule, copy ≥ 0.9	words stay readable

G.3 Word Integrity Tests

Test	Question asked
Copy	Can the model repeat a sentence containing the word?
What is X?	Free answer, labelled blindly as knowing the concept (leak), knowing another meaning or not knowing (graceful), inventing a meaning (confabulation), or not registering the word (destruction)
Context	Is an unrelated MMLU question placed after a sentence with the word still answered?
Generation	Do 20 neutral prompts per concept that use the edited words keep their answer?

Free-text answers were labelled by LLM agents that never saw which method produced them; 10%

of items were duplicated across agents, with agreement 0.92–0.97 on the exact label and 0.97–1.00 on the decisive class. Each test is also run on the real word, the word deleted, and a nonce. The *nonce floor* and the *deletion floor* are the efficacies of the unedited model on the concept questions with the edited words replaced by their nonces or deleted.

G.4 Pre-registration and Robustness

Every analysis plan was written into a pre-registration with amendments before its results existed. [**Experiments:** Robustness table: paired ΔH_3 vs. EMBER for the four CINDER variants (all 60 runs with no edit scored as no erasure; collateral gate 0.9 and 0.7; leave-one-concept-out p range); the table of development claims and their held-out verdicts; Mean and Noise baselines (development only).]

G.5 Integrity-Gated Selection

[**Experiments:** Table: runs edited, efficacy, collateral, H_3, median r, copy (exclusive / multi-meaning words) and H_4 for each method at the integrity-gated point (CINDER 0.12 and CINDER-RS 0.14 efficacy vs. EMBER 0.06).]

G.6 Direction or Token Set

Four cells cross the token set with the direction at a fixed $r = 0.5$, on the runs where EMBER has a token set; "CINDER's tokens" are CINDER's top concept-only tokens, as many as EMBER edits. All cells use dense coefficients so that every token has a coefficient on $\mathcal{F}_C$.

Efficacy / H_3	EMBER direction	CINDER direction
EMBER tokens	.20 / .39	.26 / .46
CINDER tokens	.20 / .39	.26 / .48

The direction effect (mean of the row differences) is $+0.073$ efficacy and $+0.097$ H_3, on 10 of 11 concepts (Holm p 0.097 and 0.055); the token-set effect is $+0.016$ and $+0.024$ (n.s.); the direction costs 0.02 collateral. At $r = 1$ (secondary), the cell means are .27 / .47 (EMBER tokens, EMBER direction), .27 / .45 (EMBER tokens, CINDER direction), .28 / .46 (CINDER tokens, EMBER direction) and .29 / .48 (CINDER tokens, CINDER direction); the direction effect is -0.016 efficacy ($p = 0.97$) and -0.007 H_3 ($p = 1.0$), and the token-set effect $+0.010$ and $+0.012$ (both n.s.).

G.7 The Nonce-Matching Edit

We keep the model frozen and learn a change to each edited row. The loss pushes the edited model, on concept sentences and new definitional prompts, toward what the unedited model does when the word is replaced by its nonce, while two further terms keep the word copyable (new carrier sentences) and keep the other meanings of multi-meaning tokens (sentences the model generated itself); no evaluation item is used for training. [**Experiments:** Table: runs at the nonce target, efficacy, collateral, copy, destroyed words and neutral prompts broken, for the edit inside CINDER's and EMBER's concept subspace and for free edits (seed 42, 12 held-out concepts).]

H Embedding-Space Analyses: Details

All analyses in this section use the saved factorizations of 38 concepts $\times$ 5 seeds on Gemma-2-2B-it: EMBER's matrix with row sparsity (EMBER), the sharpened matrix with row sparsity (CS), and the sharpened matrix with global sparsity (CS+GS). Unless stated otherwise, the concept feature of a run is identified without the judge, as the feature whose words match the concept's lexicon most often.

H.1 Where the Concept's Energy Lies

Matrices and fits Raw $E^{\top}_{\mathcal{V}^*}$; raw with $\hat{\mu}_B$ projected out; centred at μ_B; whitened only ($\gamma = 0$); and CS. Row-sparse fits, 38 concepts $\times$ 5 seeds; the 570 fits for the three intermediate matrices are new. A run "finds" the concept when its best feature has at least 5 lexicon words among its first 20 source words; no LLM judge is used.

Frequency Over 6.2M tokens of English Wikipedia, concept-only tokens are rarer than neutral-only ones in 35 of 38 concepts (median 9.4 vs. 12.9 per million), but their norms are not smaller (raw energy 0.9% larger). Across the vocabulary, a token's projection on $\hat{\mu}_B$ falls monotonically with its frequency (Spearman -0.69 within a concept), and rare tokens put a smaller share of their centred energy on the top background directions (0.07 vs. 0.65 at the frequent end). The tokens a raw concept feature collects are the rarest in $\mathcal{V}^*$ (median 2.5 vs. 23 per million).

Selection-free directions The concept-minus-neutral mean direction, the top generalized eigen-

vector of concept vs. neutral second moments, and the top direction of the lexicon tokens do not rise in energy rank with CS (e.g., lexicon direction rank 2 raw, which reflects $\hat{\mu}_B$; 9 without $\hat{\mu}_B$; 16.5 with CS). What sharpening changes is the energy of the lexicon direction relative to whitening alone (3.1× vs. 11.4× an average direction) and the value of the concept's cluster (App. B).

H.2 Feature Geometry and Norms

Setup 570 saved fits (EMBER, CS with row sparsity, CINDER), directions sign-aligned and compared in four metrics: the fit's own space, the raw embedding space (sharpened directions mapped back), the sharpened space (raw directions mapped forward), and the raw space with $\hat{\mu}_B$ projected out of every direction. Semantic relatedness of two features is the cosine of a small sentence encoder (all-MiniLM-L6-v2) between their source-word lists, independent of Gemma; token frequencies come from 32.6M tokens of Wikipedia.

Orthogonality Mean $|\cos|$ between directions:

Metric	EMBER	CS+RS	CINDER
Own fit space	.164	.052	.051
Raw embedding space	.164	.291	.258
Raw, $\hat{\mu}_B$ projected out	.079	.086	.079

Mapping sharpened directions back multiplies them by $T^{-1} \propto \Sigma_B^{1/2}$, which re-inflates the dominant background direction; $\hat{\mu}_B$ has cosine 0.82 with the top background principal component.

Neighbours and meaning The concept feature's nearest-neighbour angle is 50.6° (EMBER), 73.6° (CS) and 74.3° (CINDER); EMBER's neighbour is a recurring feature of rare names from the neutral sentences (*Fulda*, *Amiens*, *Ferdinando*) at the 59th relatedness percentile, the sharpened neighbours at the 92–94th. Over all pairs, the rank correlation of cosine with relatedness is higher for EMBER (0.33 vs. 0.24), but EMBER's raw cosines largely track word frequency; controlling for it, CS is higher (0.34 vs. 0.25), and among the top 0.5% of pairs by cosine only the sharpened dictionaries are semantic (75–79th vs. 51st percentile). Mapped back to the raw space, sharpened cosines lose this property: it belongs to the metric in which they were found.

Norms The concept feature's norm is at the 82nd percentile under EMBER and the 57–61st under CS; 45% of its squared norm lies along $\hat{\mu}_B$ (typical feature 12%), and with $\hat{\mu}_B$ removed it falls to the 63rd. A token's coordinate along μ_B falls from 1.1 (rarest decile) to 0 (most frequent), so features built from rare tokens with same-sign coefficients collect about $\sqrt{w}\,\|\mu_B\|$ of norm. Neither norm measures concept strength: with unit-norm coefficient rows, $\|\mathbf{z}_i\|$ is the scale of the support's vectors in the fit's metric.

Splits Two or more lexicon-matching features occur in 4–12% of runs (15–28% by the judge). Blind labels of the 56 split pairs give 52 distinct sub-topics, 2 cuts of one concept and 2 off-topic pairs (e.g., batting vs. pitching, card games vs. betting). Ancient Rome has a Rome feature and an empire feature in all 15 runs; under GS the Rome feature grows to 92–109 tokens and absorbs Roman persons, whereas EMBER's fixed 48 slots pad it with other capitals (Milan, Paris, Ottawa).

H.3 Feature Widths and Tails

See App. C.

H.4 Abstract Concepts and Contamination

Abstract concepts 13 concepts that sit between broader topics (Romanticism, Utilitarianism, Baroque, Gothic architecture, Surrealism, Art Nouveau, Neoclassicism, Gnosticism, Existentialism at $|\mathcal{S}_C| = 300$; Stoicism, Epicureanism, Impressionism, Neoplatonism, whose pages are shorter, at $|\mathcal{S}_C| = 190$) and 4 entity controls (Harry Potter, Baseball, Uranium, Golf, at both sizes), 5 seeds, EMBER, CS and CS+GS (615 fits). Each of the top 10 features by ρ was shown, blind to condition, as its 20 strongest tokens with source words and labelled specific (P), broader (B) or unrelated (N) by two independent LLM annotators (agreement 0.94, $\kappa = 0.88$; parent topics fixed before any fit). At $|\mathcal{S}_C| = 300$, a P feature is among the top 3 in 7% of abstract and 50% of entity runs under EMBER (Mann–Whitney $p = 0.043$), and CS changes top-10 presence by +0.04 (n.s.); CS+GS gains +0.22 from two concepts (Baroque, Utilitarianism). Stoicism illustrates the pattern: its top features are emotions, philosophy and religion, ethics and discourse words, and its most Stoic feature is a spelling feature that mixes *Stoic* and *stoa* with *stock* and *store*.

Contamination For Harry Potter, Baseball and Ancient Rome, a fraction $f \in \{0.1, 0.25, 0.5\}$ of the 300 concept sentences is replaced by sentences from adjacent pages (Harry Potter: Magic,

The Lord of the Rings, The Chronicles of Narnia, Fantasy literature; Baseball: Softball, Cricket; Ancient Rome: Ancient Greece, Byzantine Empire), or, as a control added afterwards, from unrelated pages. Sentences naming the concept are dropped, and the sharpening target stays the concept's own page. EMBER's mean rank of the best P feature rises from 4.1 to 6.9 at $f = 0.5$, equally with unrelated sentences (5.9 vs. 5.7, $p = 0.96$); CS keeps it in the top 10 in 80–93% of runs. The share of the feature's top tokens that come only from the inserted sentences grows with f (EMBER 0.14 / 0.20 / 0.25; CS 0.05 / 0.13 / 0.14); sharpening toward the contaminated sentences instead of the clean page raises it to 0.23 at $f = 0.5$.

H.5 Extrapolation Beyond the Sample

Benchmark 17 concepts with at least 80 page sentences outside the 300 sampled ones and a concept feature for both EMBER and CINDER in some seed; on average 734 new concept-only token types per concept, plus new neutral token types from a 1,000-page Wikipedia sample (balanced: as many new neutral as new concept-only tokens; natural: one neutral sentence per new page sentence). Three saved dictionaries, kept fixed (EMBER; CS with row sparsity; CINDER), 5 seeds. Ground truth: blind labels as concept-specific (S), adjacent (A) or unrelated (U) on 5,196 tokens, written by LLM agents that saw only the concept, the token, its source words and a short context; agreement on 300 double-labelled items 0.90 ($\kappa = 0.82$).

Rules Scores: projection on the concept feature, cosine, the coefficient of the joint dense least-squares solve, and the code share $y_{ft}|y_{ft}|/\|\mathbf{y}_{\cdot t}\|^2$. Admission: the fit's own sparsity rule over the extended set (1%), the saved absolute width, an in-sample cut-off ($|y_{ft}|$ at least the weakest coefficient of the saved support), the token's dominant feature, and share ≥ 0.5.

Results Top-10 precision among new concept-only tokens is 0.18–0.27 for every dictionary and score (random 0.065); counting adjacent tokens it is 0.49–0.69, and no score separates concept-specific from adjacent tokens (AUC 0.40–0.48). The 1% re-solve admits 15–22 tokens per run at precision 0.22–0.25, about 37% of them unrelated; fixed width and cut-off give the same precision, but fixing the width pushes out 13–15% of the original support, and Harry Potter's width grows from 48 to 59 under the 1% rule. Routing a token's shared meaning to other features (dominant feature or code share) does not help. CINDER's wider features admit more concept-specific tokens per run at equal precision (6.1 vs. 3.9), and the sharpened dictionaries rank lexicon words higher (AUC 0.91 vs. 0.67 with projection). Precision rises with a token's page frequency, from 0.28–0.30 at one occurrence to 0.59–0.71 at ten or more, for in-sample and extrapolated tokens alike. We recommend ranking by the joint solve, admitting with the in-sample cut-off, and preferring CINDER's dictionary. [**Experiments:** Figure: precision@k per score and dictionary; precision by page frequency (`results/paper_analysis/extrapolation/figures/`).]

I Qualitative Examples

The Harry Potter feature, before and after sharpening Table 6 lists the 48 tokens of the Harry Potter feature (1% row sparsity, seed 42) on EMBER's raw matrix and on the sharpened matrix, by decreasing coefficient, written as the words they come from. It is taken from our exploratory notebook fits, which differ slightly from the saved fits of Figure 5 (there, ranks 20–30 of the raw feature hold 3 Harry Potter tokens, and the sharpened feature has no neutral token). Both features open with 19 unmistakable Harry Potter tokens. After that, the raw feature drifts into unrelated names, places and taxonomic suffixes, while the sharpened feature continues into the fictional world (Gringotts, Hogsmeade, Quidditch, McGonagall, Hagrid) and is more consistent across fragments of the same word (*Gry*, *Hog*, *warts*).

Judge rejections [**Experiments:** The descriptions and confidences of TJ's wrong rejections (Valentine's Day, Heroin, World War II; all at confidence 0.85), with the feature's top tokens.]

J Reproducibility and Cost

Code and results All code, pre-registrations and result files are in our repository.[Paths for the experiments pass; shorten for the final version.]

- Single-language fits and features: `experiments/varwidth/` (fitting `core.py`, data `prep.py`, selection `select.py`); sharpening `experiments/whitening/sharpen.py`.
- Collateral-aware evaluation: `experiments/varwidth/erase_v2.py`, `collateral.py`,

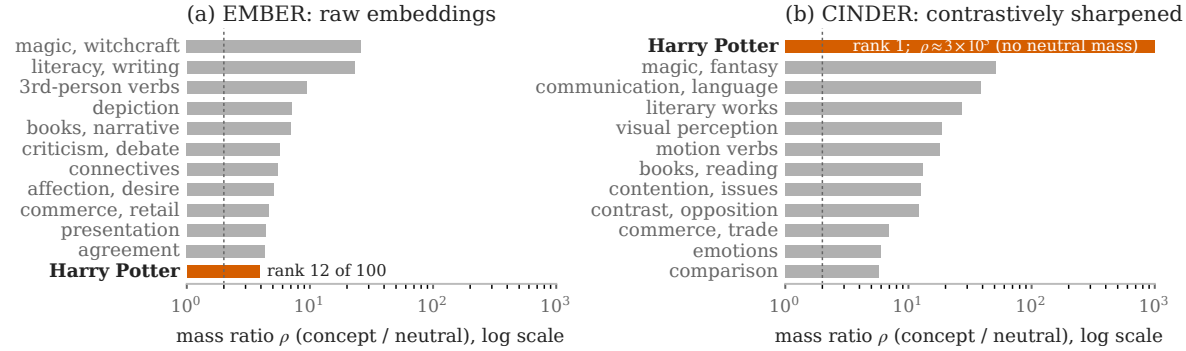


Figure 5: The top 12 of 100 sparse-MF features for Harry Potter on Gemma-2-2B-it at seed 42, ranked by EMBER's mass ratio ρ (Eq. 1); labels paraphrase each feature's LLM description. (a) Raw embeddings: eleven generic topics outrank the concept feature. (b) Sharpened matrix, same 1% row sparsity: the concept feature ranks first and has no neutral token. The raw rank varies with the seed (12, 1, 28, 48, 1 at seeds 42–46), while sharpening ranks the concept first at all five; across concepts, the raw rank is usually already near the top (§6). The dotted line marks $\tau = 2$.

`integrity.py`, `last_mile.py`, `nonce_edit.py`, `confirm_stats.py`; pre-registration `results/varwidth/layers/PREREG.md`; results in `results/varwidth/layers/collateral/` (`confirm.md`, `last_mile.md`, `twobytwo.md`, `nonce_edit.md`).

- Feature-level ladder and judges: `results/varwidth/layers/SUMMARY.md`, `experiments/varwidth/j3_eval.py`.
- Multilingual study (branch `orip-ml`): `experiments/ml/`, `results/ml/PREREG_RESULTS.md`, `results/ml/UNION_RESULTS.md`.
- Embedding-space analyses: `experiments/paper_analysis/`, `results/paper_analysis/`.
- Figures: `paper/figures/src/`.

Compute and cost [**Experiments:** GPU types and hours (NVIDIA TITAN Xp, RTX 2080 Ti and others on the university cluster); Gemini spend for feature judging, lexicons and question generation (about 16 NIS for the single-language ladder and about 17 NIS for the multilingual study).]

Ranks	EMBER (raw)	CINDER (sharpened)
1–19	all Harry Potter (Gryffindor, Dumbledore, Slytherin, Rowling, Voldemort, Weasley, . . .)	all Harry Potter (Gryffindor, Slytherin, Dumbledore, Weasley, Hogwarts, . . .)
20–30	Kipling, **Ginny**, **Lupin**, Beatles, Arrondissement, Escondido, vice-chancellor, Cerambycidae, Alamos, Sherlock, Balompié	**Hogwarts**, **Gryffindor**, **Ginny**, **Hogsmeade**, **Gringotts**, **Hogwarts**, **Lupin**, **Remus**, **Granger**, **Slytherin**, **JK**
31–48	**Remus**, Chrysomelidae, Trump, Nestlé, Bloomsbury, tesla, Fiennes, Polygyridae, Cerambycidae, Picasso, Assassins, Steelers, **Gryffindor**, Chhattisgarh, Lego, Kinect, Reagan, HBO	**Sirius**, **wand**, **McGonagall**, **Muggle**, **Ron**, **Goblet**, Sherlock, **Hagrid**, **Quidditch**, **Beauxbatons**, **potions**, **Rowling**, **Riddle**, **Grindelwald**, **Quirrell**, Holmes, **Merlin**, NHP

Table 6: Tokens of the Harry Potter feature in EMBER's raw matrix and in the sharpened matrix (seed 42), shown as their source words. Bold: clearly Harry Potter tokens beyond rank 19. Ranks 20–30 hold 2 of 11 Harry Potter tokens before sharpening and 11 of 11 after.